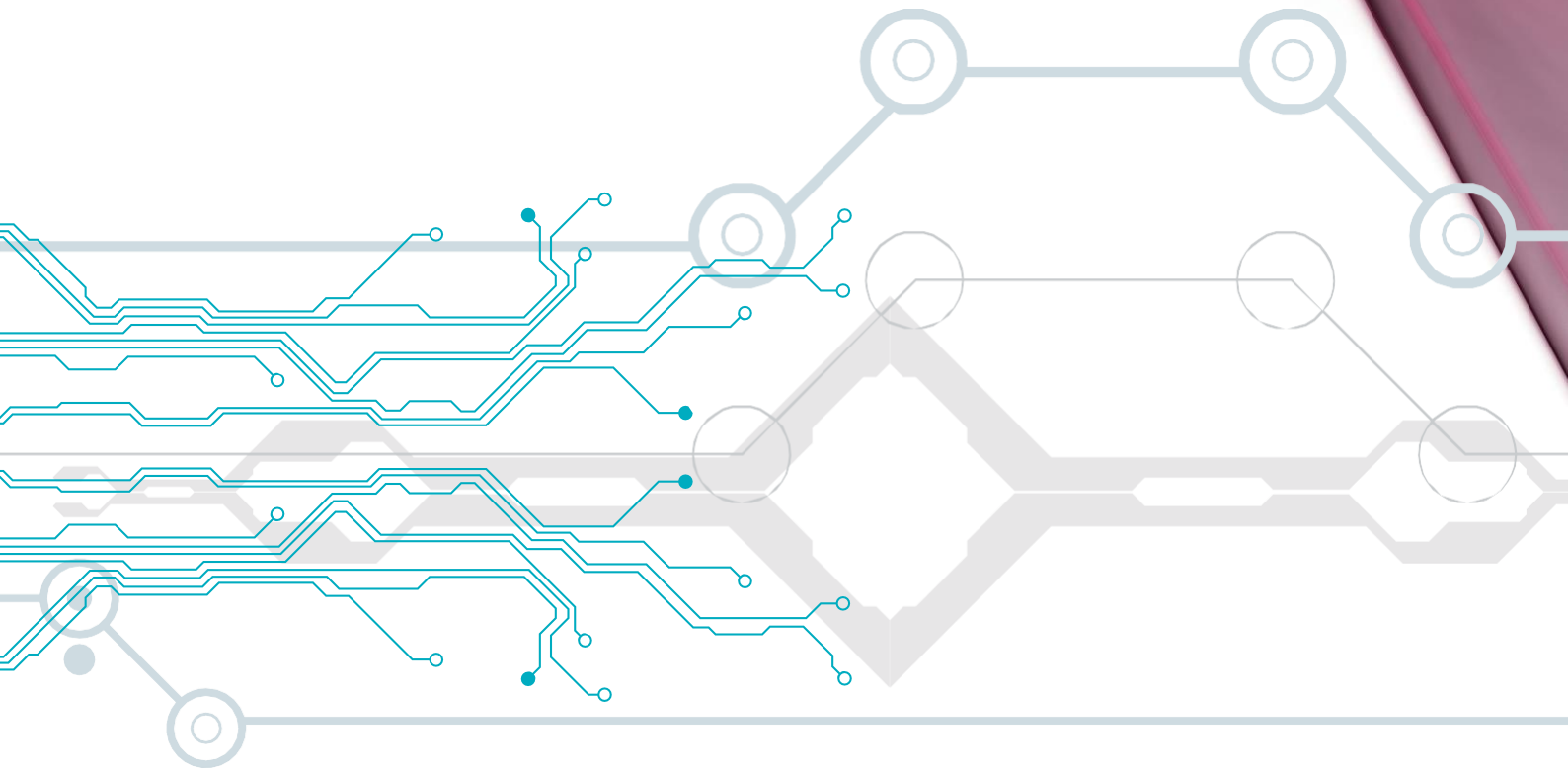
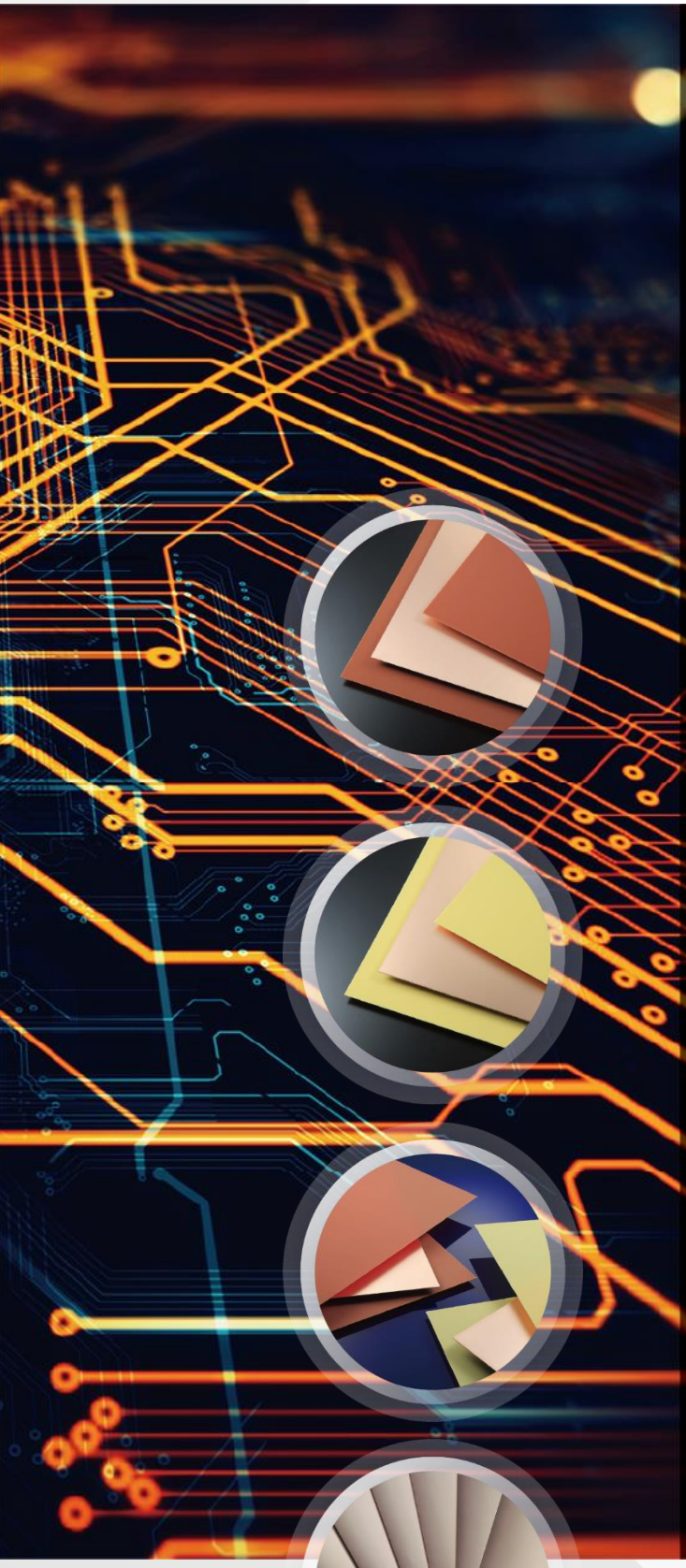
COPPER CLAD LAMINATES



RESONAC



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TYPE-F: Superfine flat technology

Recommended PWB Materials by Application

	Product	MCL-E-795G	MCL-E-705G	MCL-E-700G (Type R)	MCL-E-770G (Type R)	TYPE-F GEA705G/ GEA-770G/ GEA-795G	MCL-E-679FG	MCL-HS200	MCL-HS100	TYPE-F GH-100 (Type D)/ GH-200 (Type D)	MCL-HD60/ HD80/HD90
	Feature	High Tg, High Elastic Modulus, Low CTE	High Tg, High Elastic Modulus, Low CTE	High Tg, High Elastic Modulus, Low CTE	High Tg, High Elastic Modulus, Low CTE	Thin Prepreg with Superior Surface Smoothness	High Elastic Modulus, Low CTE	High Tg, Low Transmission Loss, Low CTE	High Tg, Low Transmission Loss, Low CTE	Low Transmission Loss, Thin Prepreg with Superior Surface Smoothness	High Dk
Environmentally Friendly	Halogen Free	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
	Lead Free Application (288℃)	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
Semiconductor Packages	Logic	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
	Memory	◎	◎	◎	◎	◎	◎	○	○	○	○
	SiP*1	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
	High-speed Modules	○	○	○	○	○	–	◎	◎	◎	◎
ICT Infrastructure	Router• Server	–	–	–	–	–	–	○	○	○	○
	Antenna	–	–	–	–	–	–	○	○	○	◎
Automotive	Control Unit	◎	◎	◎	◎	◎	◎	–	–	–	–
	Radars/ Sensors	–	–	–	–	–	–	◎	◎	◎	
	Power Semi-conductors	◎	◎	◎	–	–	–	–	–	–	–
	ECU	–	–	–	–	–	◎	–	–	–	–

*1: System in Package

	Product	MCF-200HS	MCF-770G (Type P)	MCL-LW-990 (Type RFD)	MCL-LW-900G/910G	MCL-HE-679G (Type S)	MCL-I-671	TD-002	PF-EL
	Feature	Low Transmission Loss, Low CTE	Low CTE Adhesive Film with Primer	High Tg, Low Transmission Loss	High Tg, Low Transmission Loss	High Tg, Mid Loss	Polyimide material	Reduction of Solder Crack, Low Elastic Modulus	Copper Foil for Fine Patterning
Environmentally Friendly	Halogen Free	◎	◎	-	◎	◎	-	-	◎
	Lead Free Application（288℃）	◎	◎	◎	◎	◎	◎	◎	◎
Semiconductor Packages	Logic	◎	◎	-	-	-	-	-	◎
	Memory	○	○	-	○	◎	-	-	○
	SiP*1	◎	◎	-	-	-	-	-	○
	High-speed Modules	◎	○	◎	◎	◎	-	-	○
ICT Infrastructure	Router• Server	○	-	○	◎	◎	◎	-	-
	Antenna	○	-	◎	◎	-	-	-	-
Automotive	Control Unit	-	○	-	-	-	-	◎	○
	Radars/ Sensors	-	-	◎	○	○	-	◎	-
	Power Semi-conductors	-	-	-	-	-	-	◎	-
	ECU	-	-	-	-	-	-	◎	-

General Characteristics

Product	Property	Unit	MCL-E-795G	MCL-E-795G (Type LH)	MCL-E-705G	MCL-E-705G (Type L)	MCL-E-705G (Type LH)	MCL-E-700G (Type R)	MCL-E-770G (Type R)	MCL-E-770G (Type RLH)
Electrical (SPDR)	Thickness	mm	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	Glass Cloth	-	E-glass	S-glass(H.D.)	E-glass	S-glass	S-glass(H.D.)	E-glass	E-glass	S-glass(H.D.)
	Dk (1 GHz)	-	4.3 - 4.5	4.1 - 4.3	4.4 - 4.6	4.1 - 4.3	4.1 - 4.3	4.7 - 4.9	4.3 - 4.5	4.0 - 4.2
	Dk (10 GHz)		4.3 - 4.5	4.1 - 4.3	4.4 - 4.6	4.1 - 4.3	4.1 - 4.3	4.6 - 4.8	4.2 - 4.4	3.9 - 4.1
	Df (1 GHz)	-	0.005 - 0.007	0.005 - 0.007	0.007 - 0.009	0.007 - 0.009	0.007 - 0.009	0.008 - 0.010	0.005 - 0.007	0.005 - 0.007
	Df (10 GHz)		0.006 - 0.008	0.006 - 0.008	0.009 - 0.011	0.009 - 0.011	0.009 - 0.011	0.008 - 0.010	0.006 - 0.008	0.006 - 0.008
Physical	Peel Strength (12μm)	kN/m	0.8 - 1.0	0.8 - 1.0	0.8 - 1.0	0.8 - 1.0	0.8 - 1.0	0.9 - 1.1	0.7 - 0.9	0.7 - 0.9
	Peel strength (18μm)	kN/m	0.9 - 1.1	0.9 - 1.1	0.9 - 1.1	0.9 - 1.1	0.9 - 1.1	1.0 - 1.2	0.8 - 1.0	0.8 - 1.0
	Tensile modulus X*4	GPa	26 - 28	31 - 33	20 - 22	22 - 24	23 - 25	20 - 22	18 - 20	20 - 22
	Flexural modulus X*4	GPa	35 - 37	40 - 42	32 - 34	34 - 36	37 - 39	32 - 34	30 - 32	34 - 36
	Tg (TMA)	℃	260 - 290	260 - 290	250 - 270	250 - 270	250 - 270	250 - 270	260 - 280	260 - 280
	Tg (DMA)		315 - 345	315 - 345	295 - 305	295 - 305	295 - 305	295 - 305	300 - 330	300 - 330
	CTE x (<Tg) (Comp.)	ppm/℃	7 - 9	4.5 - 6.5	8.5 - 10.5	6 - 8	6 - 8	11 - 13	7 - 9	5 - 7
	CTE y (<Tg) (Comp.)		7 - 9	4.5 - 6.5	8.5 - 10.5	6 - 8	6 - 8	11 - 13	7 - 9	5 - 7
	CTE z (<Tg) (Comp.)*4		10 - 15	10 - 15	10 - 15	10 - 15	10 - 15	15 - 25	10 - 15	10 - 15
	CTE z (>Tg) (Comp.)*4		70 - 100	70 - 100	70 - 100	70 - 100	70 - 100	90 - 120	70 - 100	70 - 100
	CTE z (<Tg)(DIC)*4		8.5	5.5	9.6	7.0	6.9	13.1	-	-
	Thermal conductivity	W/(m·K)	0.55 - 0.75	0.55 - 0.75	0.55 - 0.75	0.55 - 0.75	0.55 - 0.75	0.60 - 0.80	0.55 - 0.75	0.55 - 0.75
Slash sheet No.	IPC-4101		42	N/A (S-glass)	42	N/A (S-glass)	N/A (S-glass)	42	42	N/A (S-glass)

*1: Cavity resonator; *2: RT foil; *3: HVLP foil, *4: 0.8 mmt; *5: Tensile

Product	Property	Unit	TYPE-F GEA-705G	TYPE-F GEA-770G	TYPE-F GEA-795G	MCL-E-679FG (Type R)	MCL-E-679FG (Type S)	MCL-HS200	MCL-HS200 (Type D)	MCL-HS100
Electrical (SPDR)	Thickness	mm	0.025	0.025	0.020	0.2	0.2	0.2	0.2	0.2
	Glass Cloth	-	E-glass	E-glass	S-glass	E-glass	E-glass	E-glass	Low Dk-glass	E-glass
	Dk (1 GHz)	-	3.7 - 4.0	3.6 - 3.9	3.7 - 3.9	4.9 - 5.2	4.9 - 5.2	3.8 - 4.0	3.4 - 3.6	4.2 - 4.4
	Dk (10 GHz)		3.7 - 4.0	3.6 - 3.9	3.6 - 3.8	4.8 - 5.1	4.8 - 5.1	3.8 - 4.0	3.4 - 3.6	4.2 - 4.4
	Df (1 GHz)	-	0.008 - 0.010	0.007 - 0.009	0.004 - 0.006	0.0130 - 0.0150	0.0130 - 0.0150	0.0030 - 0.0040	0.0020 - 0.0030	0.0030 - 0.0040
	Df (10 GHz)		0.008 - 0.010	0.007 - 0.009	0.005 - 0.007	0.0140 - 0.0160	0.0140 - 0.0160	0.0040 - 0.0050	0.0020 - 0.0030	0.0050 - 0.0060
Physical	Peel Strength (12μm)	kN/m	0.7 - 0.9	0.5 - 0.7	0.5 - 0.7	0.8 - 1.0	0.9 - 1.1	0.5 - 0.7	0.5 - 0.7	0.5 - 0.7
	Peel strength (18μm)	kN/m	-	-	—	0.9 - 1.1	1.0 - 1.2	0.6 - 0.8	0.6 - 0.8	0.5 - 0.7
	Tensile modulus X*4	GPa	17 - 19	18 - 20	20 - 22	20 - 26	21 - 27	17 - 20	17 - 20	20 - 22
	Flexural modulus X*4	GPa	-	-	—	23 - 28	24 - 29	21 - 26	21 - 26	23 - 28
	Tg (TMA)	℃	250 - 270	260 - 280	260 - 290	165 - 175	175 - 185	250 - 270	250 - 270	240 - 260
	Tg (DMA)		295 - 305	280 - 300	330 - 350	200 - 220	210 - 230	330 - 350	330 - 350	240 - 260
	CTE x (<Tg) (Comp.)	ppm/℃	8 - 10*5	6 - 8*5	5 - 7*5	13 - 15	12 - 14	8 - 10	8 - 10	6 - 8
	CTE y (<Tg) (Comp.)		8 - 10*5	6 - 8*5	5 - 7*5	13 - 15	12 - 14	8 - 10	8 - 10	6 - 8
	CTE z (<Tg) (Comp.)*4		15 - 25	15 - 25	14 – 24	23 - 33	20 - 30	25 - 35	25 - 35	20 - 30
	CTE z (>Tg) (Comp.)*4		70 - 120	90 - 120	140 - 170	140 - 170	130 - 160	140 - 200	140 - 200	130 - 180
	CTE z (<Tg)(DIC)*4		-	-	—	-	-	-	-	-
	Thermal conductivity	W/ (m·K)	-	0.40 - 0.60	0.55 - 0.75	0.79 - 0.83	0.79 - 0.83	0.60 - 0.80	0.60 - 0.80	0.40 - 0.50
Slash sheet No.	IPC-4101		42	42	42	128,130	128,130	44	N/A (Low Dk-glass)	44

General Characteristics

Product	Property	Unit	MCL-HS100 (Type D)	GH-200 (Type D)	GH-100 (Type D)	MCL-HD60	MCL-HD60 (Type D)	MCL-HD80	MCL-HD90	MCL-HD90 (Type D)	MCF-200HS	MCF-770G (Type P)
Electrical (SPDR)	Thickness	mm	0.2	0.025	0.025	0.2	0.2	0.2	0.2	0.2	0.025	0.025
	Glass Cloth	-	Low Dk-glass	Low Dk-glass	Low Dk-glass	E-glass	Low Dk-glass	E-glass	E-glass	Low Dk-glass	-	-
	Dk (1 GHz)	-	3.7 - 3.9	3.2 - 3.4	3.3 - 3.5	-	-	-	-	-	-	-
	Dk (10 GHz)		3.5 - 3.7	3.2 - 3.4	3.3 - 3.5	6.0 - 6.2	5.8 - 6.0	8.0 - 8.2	9.0 - 9.2	8.2 - 8.4	2.9 - 3.1	3.2 - 3.3
	Df (1 GHz)	-	0.0020 - 0.0030	0.001 - 0.003	0.002 - 0.004	-	-	-	-	-	-	-
	Df (10 GHz)		0.0025 - 0.0035	0.0020 - 0.0040	0.0030 - 0.0050	0.0050 - 0.0060	0.0040 – 0.0050	0.0050 - 0.0060	0.0040 – 0.0050	0.0030 - 0.0040	0.0030 - 0.0040	0.0060 - 0.0080
Physical	Peel Strength (12μm)	kN/m	0.5 - 0.7	0.4 - 0.6	0.4 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.7	0.9 - 1.0
	Peel strength (18μm)	kN/m	0.5 - 0.7	-	-	0.6 - 0.7	0.6 - 0.7	0.6 - 0.7	0.6 - 0.7	0.6 - 0.7	-	-
	Tensile modulus X*4	GPa	20 - 22	10 - 12	10 - 12	19 - 23	18 - 22	18 - 22	18 - 22	17 - 21	5 - 7	8 -10
	Flexural modulus X*4	GPa	23 - 28	-	-	-	-	-	-	-	-	-
	Tg (TMA)	℃	240 - 260	220 - 240	240 - 260	-	-	-	-	-	-	-
	Tg (DMA)		240 - 260	250 - 270	240 - 260	230 - 240	230 - 240	230 - 240	230 - 240	230 - 240	250 - 270	290 - 310
	CTE x (<Tg) (Comp.)	ppm/℃	6 - 8*5	13 - 15*5	13 - 15*5	9 - 11*5	9 - 11*5	10 - 12*5	10 - 12*5	10 - 12*5	35 - 40*5	18 - 20*5
	CTE y (<Tg) (Comp.)		6 - 8*5	13 - 15*5	13 - 15*5	9 - 11*5	9 - 11*5	10 - 12*5	10 - 12*5	10 - 12*5	35 - 40*5	18 - 20*5
	CTE z (<Tg) (Comp.)*4		20 - 30	30 - 50	30 - 50	-	-	-	-	-	-	-
	CTE z (>Tg) (Comp.)*4		130 - 180	210 - 240	210 - 240	-	-	-	-	-	-	-
	CTE z (<Tg)(DIC)*4		-	-	-	-	-	-	-	-	-	-
	Thermal conductivity	W/(m·K)	0.40 - 0.50	0.50 - 0.70	-	-	0.60 - 0.80	0.60 - 0.80	0.60 - 0.80	0.60 - 0.80	-	-
Slash sheet No.	IPC-4101		N/A (Low Dk-glass)	N/A (Low Dk-glass)	N/A (Low Dk-glass)	44	N/A (Low Dk-glass)	44	44	N/A (Low Dk-glass)	N/A	N/A

*1: Cavity resonator; *2: RT foil; *3: HVLP foil, *4: 0.8 mmt; *5: Tensile

Product	Property	Unit	MCL-LW-990 (Type RFD)	MCL-LW-910G	MCL-LW-900G	MCL-HE-679G (Type S)	MCL-I-671	TD-002
Electrical (SPDR)	Thickness	mm	0.8	0.8	0.8	0.8	0.2	0.4
	Glass Cloth	-	Low Dk-glass	Low Dk-glass	E-glass	E-glass	E-glass	E-glass
	Dk (1 GHz)	-	-	-	-	3.7 - 3.9[*1]	4.2 - 4.3	3.6 - 3.8[*1]
	Dk (10 GHz)		3.1 - 3.3[*1]	3.2 - 3.4[*1]	3.6 - 3.8[*1]	3.6 - 3.8[*1]	-	-
	Df (1 GHz)	-	-	-	-	0.006 - 0.008*1	0.013 - 0.015	0.011 - 0.013[*1]
	Df (10 GHz)		0.0018 - 0.0028[*1]	0.0025 - 0.0035[*1]	0.0040 - 0.0050[*1]	0.008 - 0.010[*1]	-	-
Physical	Peel Strength (12μm)	kN/m	-	-	-	-	-	-
	Peel strength (18μm)	kN/m	0.4 - 0.6[*3]	0.4 - 0.7[*3]	0.4 - 0.7[*2]	0.5 - 0.7[*2]	1.3 - 1.5	0.8 - 0.9
	Tensile modulus X[*4]	GPa	16 - 22	18 - 24	18 - 24	17 - 23	21 - 24	6 - 9
	Flexural modulus X[*4]	GPa	14 - 19	16 - 21	16 - 21	23 - 26	24 - 26	5 - 8
	Tg (TMA)	℃	180 - 230	190 - 210	190 - 210	180 - 190	200 - 213	155 - 170
	Tg (DMA)		220 - 260	220 - 260	220 - 260	260 - 280	230 - 245	180 - 200
	CTE x (<Tg) (Comp.)	ppm/℃	7 - 10	12 - 15	12 - 15	12 - 15	12 - 15	6 - 9[*5]
	CTE y (<Tg) (Comp.)		7 - 10	12 - 15	12 - 15	12 - 15	12 - 16	6 - 9[*5]
	CTE z (<Tg) (Comp.)[*4]		35 - 55	35 - 45	35 - 45	30 - 40	50 - 80	80 - 130
	CTE z (>Tg) (Comp.)[*4]		240 - 290	230 - 280	230 - 280	190 - 230	200 - 300	200 - 300
	CTE z (<Tg)(DIC)[*4]		-	-	-	-	-	-
	Thermal conductivity	W/ (m·K)	0.30 - 0.50	0.40 - 0.60	0.40 - 0.60	0.40 - 0.50	0.30 - 0.40	-
Slash sheet No.	IPC-4101		N/A （Low Dk-glass）	N/A （Low Dk-glass）	44	42	42	N/A

Halogen Free, High Tg, High Elastic Modulus, Low CTE Multilayer Material

MCL-E-795G

Product

- **CCL MCL-E-795G**
- **Prepreg GEA-795G**

Applications

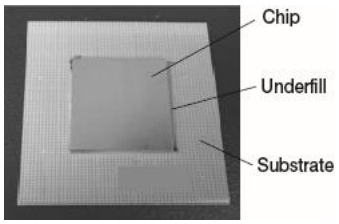
- **Semiconductor packages（FC-BGA）**
- **Center core for Buildup construction**

Features

- MCL-E-795G has low CTE value in X, Y directions and can reduce warpage of package significantly.
- MCL-E-795G(Type LH) has low CTE value.
- Well-suited for center core of buildup construction.

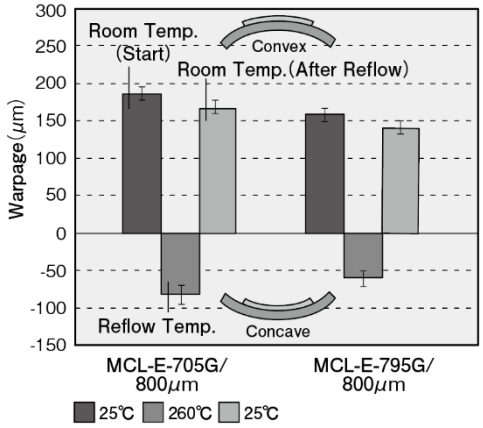
Characteristics

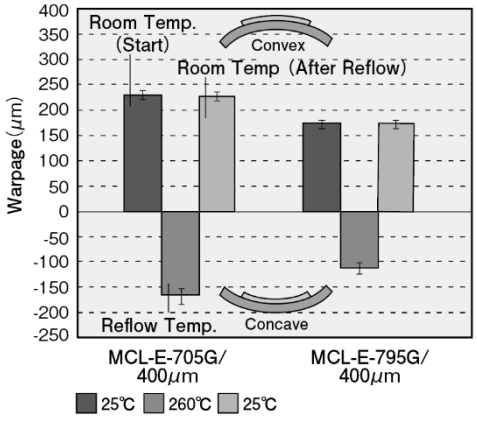
- **Warpage Performance [3-2-3 construction]**

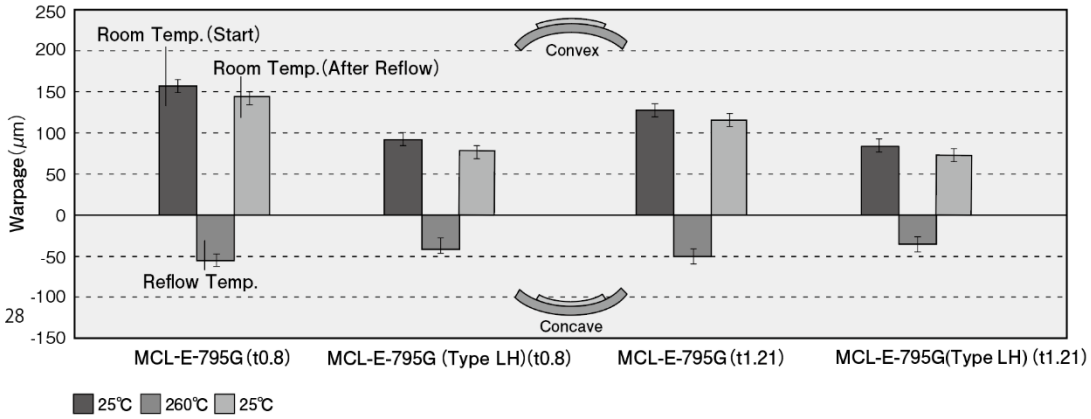


TEG Substrate
- ＊Package size: 40 mm x 40 mm
- ＊Core thickness: 0.4 mm / 0.8 mm
- ＊BU thickness: 20 μm
- ＊SR thickness: 19 μm

TEG Chip
- ＊Die size: 20 mm x 20 mm
- ＊Die height: 775 μm







●Copper Clad Laminate

(t 0.4 mm)

Item			Condition*4	Unit	Actual value				Reference (IPC-TM-650)
					MCL-E-795G	MCL-E-795G (Type X)	MCL-E-795G (Type L)	MCL-E-795G (Type LH)	
Tg	TMA method		A	℃	260～290				2.4.24
	DMA method		A		315～345				-
CTE*1	X(30～120℃)		A	ppm/℃	3.0 ～ 5.0	2.0～4.0	1.0～3.5	0.5～3.0	-
	Y(30～120℃)		A		3.0 ～ 5.0	2.0～4.0	1.0～3.5	0.5～3.0	
	Z*2	(＜Tg)	A		10 ～ 15				2.4.24
		(＞Tg)			70～100				
Solder Heat Resistance(260℃)			A	sec.	≧300				-
T-260(Without Copper)			A	min.	≧60				2.4.24.1
T-288(Without Copper)			A		≧60				
Decomposition Temperature (TGA method, 5% weight loss)			A	℃	470～490				2.3.4 0
Copper Peel strength	12 μm		A	kN/m	0.8～1.0				2.4.8
	18 μm		A		0.9～1.1				
Flexural Modulus(Lengthwise)*2			A	GPa	35～37	36～38	38～40	40～42	-
Dielectric Constant	1 GHz*3		A	—	4.3～4.5	4.2～4.4	4.1～4.3	4.1～4.3	-
Dissipation Factor	1 GHz*3		A	—	0.005～0.007	0.005～0.007	0.005～0.007	0.005～0.007	-
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}				2.5.17
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}				2.5.17
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}				-
			D-2/100		1×10^{12}～1×10^{15}				-

*1）Heating Rate:10℃/min. (Comp.) *2）800 μm thickness. *3）Measured by SPDR. *4）Refer to "Condition Note" *Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper foil thickness	Thickness Code	Laminate Thickness
MCL-E-795G	—	12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP, PF)	M0.06	0.06 mm
			0.1	0.11 mm
			0.2	0.21 mm
			0.41	0.41 mm
			0.61	0.61 mm
			0.81	0.82 mm
			1.01	1.02 mm
			1.21	1.23 mm
			1.41	1.43 mm
			1.61	1.64 mm
	(X)	12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP, PF)	0.41	0.41 mm
			0.61	0.62 mm
			0.71	0.72 mm
			0.81	0.82 mm
			1.01	1.02 mm
			1.21	1.23 mm
			1.41	1.43 mm
			1.61	1.64 mm
	(L)	12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP, PF)	0.1	0.11 mm
			0.2	0.21 mm
			0.41	0.41 mm
			0.61	0.62 mm
			0.71	0.72 mm
			0.81	0.82 mm
			1.01	1.03 mm
			1.21	1.24 mm
			1.41	1.44 mm
			1.61	1.65 mm
	(LH)	12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP, PF)	0.1	0.11 mm
			D0.15	0.16 mm
			0.2	0.22 mm
			0.41	0.43 mm
			0.61	0.64 mm
			0.71	0.75 mm
			0.81	0.85 mm
			1.01	1.07 mm
			1.21	1.28 mm
			1.41	1.49 mm
			1.61	1.71 mm

Note1）STD：Standard copper foil, LP：Low Profile copper foil, PF: Profile-free copper foil.
Note2）Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type			Glass Cloth Style	Properties Resin Content (%)	Properties Dielectric Thickness after Lamination*1
GEA-795G	-	0.03	(1027N76)	1027	76±2	0.044 mm
		0.04	(1037N76)	1037	76±2	0.053 mm
		0.06	(1078N66)	1078	66±2	0.072 mm
		0.1	(2116N58)	2116	58±2	0.124 mm
	(L)	0.03	(1027N73)	1027	73±2	0.039 mm
		0.03	(1027N76)	1027	76±2	0.044 mm
		0.04	(1037N76)	1037	76±2	0.053 mm
		0.06	(1078N66)	1078	66±2	0.072 mm
		0.06	(1280N63)	1280	63±2	0.072 mm
		0.08	(2013N58)	2013	58±2	0.096 mm
		0.1	(2116N58)	2116	58±2	0.125 mm

*1）The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is within 0%.

Halogen Free, High Tg, High Elastic Modulus, Low CTE Multilayer Material

MCL-E-705G

Product

- **CCL MCL-E-705G**
- **Prepreg GEA-705G**

Applications

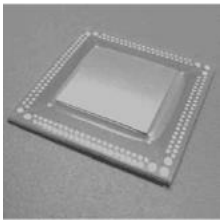
- **Semiconductor packages（FC-BGA, FC-CSP, PoP, SiP）**
- **HDI, PWB**
- **Thinner module, PWB**

Features

- MCL-E-705G has low CTE values in X,Y directions and reduces warpage of package substrate significantly.
- MCL-E-705G(Type L) and MCL-E-705G(Type LH) have lower CTE values by applying low CTE glass cloths.
- Well-suited for build-up construction.

Characteristics

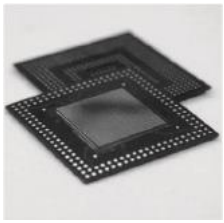
- **Warpage of FC-BGA**



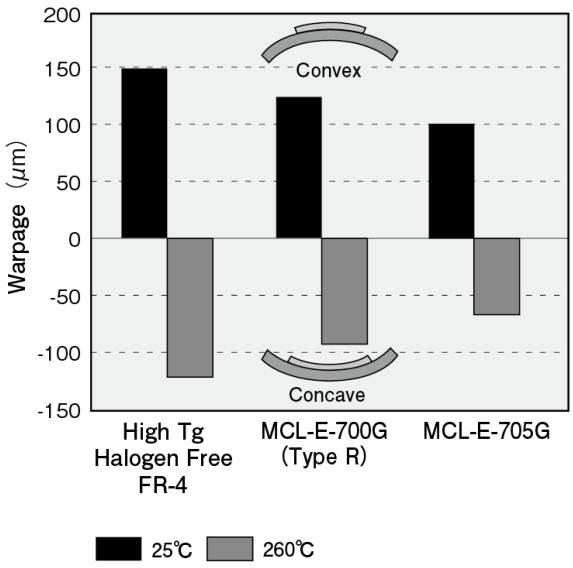
TEG Chip
- ＊Chip size: 20 mm×20 mm
- ＊Chip thickness: 0.725 mm
- ＊Bump diameter: 80 μm
- ＊Bump pitch: 200 μm

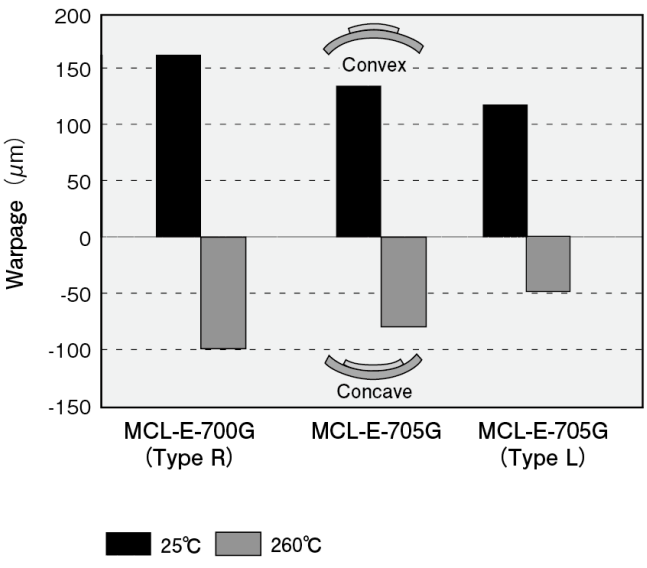
TEG Substrate
- ＊Sub size: 35 mm×35 mm
- ＊Core thickness: 0.4 mm
- ＊Build up thickness: 30 μm×2
- ＊SR thickness: 20 μm

- **Warpage of PKG**



TEG Substrate spec
- ＊Size: 14 mm×14 mm
- ＊Total thickness: 250 μm
- ＊SR thickness: 20 μm （SR-7200G）
- ＊Prepre thickness: 40 μm
- ＊Core thickness: 110 μm





●Copper Clad Laminate

(t 0.4 mm)

Item			Condition*4	Unit	Actual Value				Reference (IPC-TM-650)
					MCL-E-705G	MCL-E-705G (Type X)	MCL-E-705G (Type L)	MCL-E-705G (Type LH)	
Tg	TMA method		A	℃	250～270				2.4.24
	DMA method		A		295～305				—
CTE*1	X*2(30～100℃)		A	ppm/℃	5～7	4～6	3～4	2.5～3.5	—
	Y*2(30～100℃)		A		5～7	4～6	3～4	2.5～3.5	
	Z*5	(＜Tg)	A		10～15				2.4.24
		(＞Tg)			70～100				
Solder Heat Resistance(260℃)			A	sec.	≧300				—
T-260(Without Copper)			A	min.	≧60				2.4.24.1
T-288(Without Copper)			A		≧60				
Decomposition Temperature (TGA method 5% Weight Loss)			A	℃	430～450				2.3.40
Heat Resistance for HDI Process(Semi-Additive)			260℃ Reflow	cycles	≧20				—
Copper Peel Strength	12 μm		A	kN/m	0.8～1.0				2.4.8
	18 μm		A		0.9～1.1				
Surface Roughness(Ra)			A	μm	2～3				2.2.17
Flexural Modulus(Lengthwise)*5			A	GPa	32～34	33～35	34～36	37～39	—
Dielectric Constant	10 GHz*3		A	—	4.4～4.6	4.4～4.6	4.1～4.3	4.1～4.3	—
Dissipation Factor	10 GHz*3		A	—	0.009～0.011				—
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}				2.5.17
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}				
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}				—
			D-2/100		1×10^{12}～1×10^{14}				—

*1) Heating Rate: 10℃/min. *2)Tensile *3)Measured by SPDR. *4)Refer to"Condition Note" *5)t0.8 mm
*Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-E-705G	— (L)	2, 3, 12 μm(LP, PF)	T0.06	0.06 mm
		12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP,PF)	M0.06	0.06 mm
			0.1	0.11 mm
			M0.11	0.10 mm
			M0.15	0.16 mm
			M0.22	0.21 mm
			0.2	0.21 mm
			0.31	0.31 mm
			0.41	0.41 mm
			0.51	0.52 mm
			0.61	0.62 mm
			0.71	0.72 mm
			0.81	0.82 mm
	(X)	12 μm 18 μm (STD) 2 μm 3 μm 12 μm (LP,PF)	0.41	0.41 mm
			0.61	0.62 mm
			0.81	0.82 mm
	(LH)		M0.06	0.06 mm
			0.1	0.11 mm
			D0.15	0.16 mm
			0.2	0.21 mm
			0.26	0.26 mm

Note1) STD: Standard copper foil, LP: Low profile copper foil, PF: Profile-free copper foil.
Note2) "U" for 1-ply; "T" for 2ply.
Note3) Laminate thickness means dielectric layer thickness.

Part Number	Type		Glass Cloth Style	Properties Resin Content (%)	Properties Dielectric Thickness after Lamination *1
GEA-705G	—	≦0.025	(See P.18-19)		
		0.025	(1017N73)	1017	73±2	0.025 mm
		0.03	(1027N73)	1027	73±2	0.039 mm
		0.04	(1037N73)	1037	73±2	0.047 mm
		0.06	(1078N65)	1078	65±2	0.070 mm
		0.1	(2116N58)	2116	58±2	0.126 mm
	(L)	0.025	(L1017N73)	1017	73±2	0.025 mm
		0.03	(L1027N73)	1027	73±2	0.040 mm
		0.04	(L1037N73)	1037	73±2	0.047 mm
		0.06	(L1078N65)	1078	65±2	0.071 mm
		0.1	(L2116N58)	2116	58±2	0.127 mm
Reference (IPC-TM-650)				2.3.16	—	

*1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Value change depending on the press condition or inner layer pattern.

Halogen Free, High Tg, High Elastic Modulus, Low CTE Multilayer Material

MCL-E-700G（Type R）

Product

- **CCL MCL-E-700G (Type R)**
- **Prepreg GEA-700G**

Applications

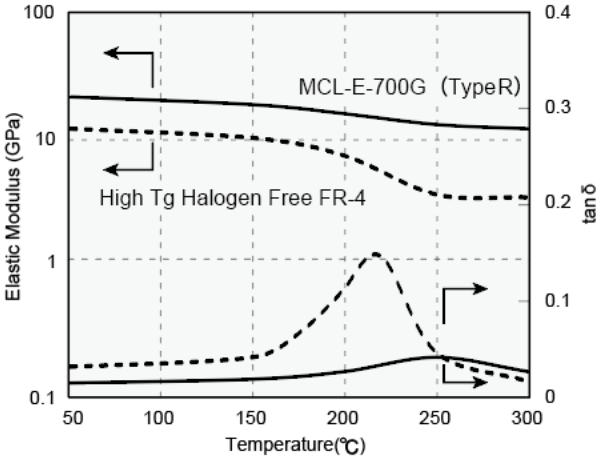
- **Semiconductor packages（FC-BGA, FC-CSP, PoP, SiP）**
- **HDI, PWB**
- **Thinner Module PWB**

Features

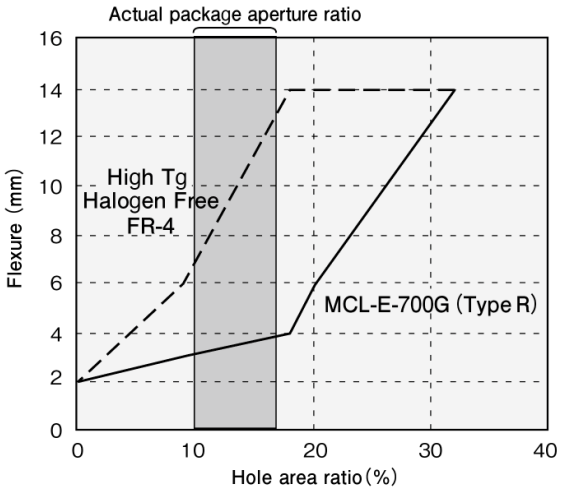
- Has low CTE values in X,Y directions and reduces warpage of package substrate significantly.
- Well-suited for build-up construction.
- Good drill processability : lower process cost.

Characteristics

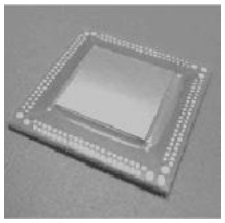
Elastic Modulus



Amount of deflection by hole area

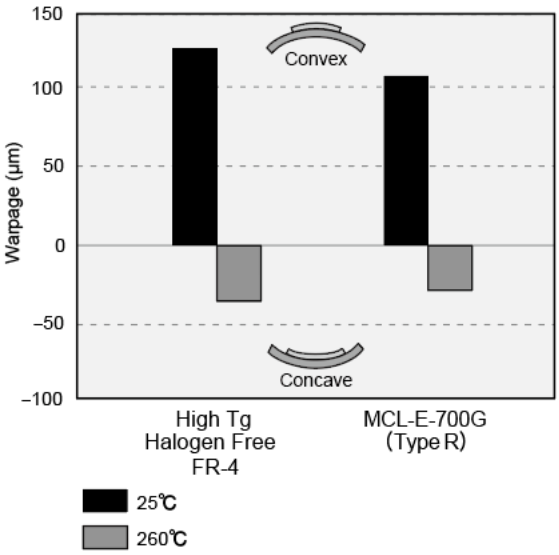


Warpage of FC-BGA



TEG Chip
- ＊Chip size : 20 mm×20 mm
- ＊Chip thickness : 0.725 mm
- ＊Bump diameter : 80 μm
- ＊Bump pitch : 200 μm

TEG Substrate
- ＊Sub size : 35 mm×35 mm
- ＊Core thikness : 0.7 mm
- ＊Build up thickness : 30 μm×2
- ＊SR thickness : 20 μm



●Copper Clad Laminate

(t0.4 mm)

Item			Condition*3	Unit	Actual Value MCL-E-700G (Type R)	Reference (IPC-TM-650)
Tg	TMA method		A	℃	250～270	2.4.24
	DMA method		A		295～305	—
CTE*1	X (30～120℃)		A	ppm/℃	8～10	—
	Y (30～120℃)				8～10	
	Z*4	(＜Tg)	A		15～25	2.4.24
		(＞Tg)	A		90～120	
Solder Heat Resistance (260℃)			A	sec.	≧300	—
T-260 (Without Copper)			A	min.	≧60	2.4.24.1
T-288 (Without Copper)					≧60	
Decomposition Temperature(TGA method 5% Weight Loss)			A	℃	430～450	2.3.40
Heat Resistance for HDI Process (Semi-Additive)			260℃ Reflow	cycle	≧20	—
Copper Peel Strength	12μm		A	kN/m	0.9～1.1	2.4.8
	18μm				1.0～1.2	
Surface Roughness(Ra)			A	μm	2～3	2.2.17
Flexural Modulus (Lengthwise) *4			A	GPa	32～34	—
Dielectric Constant	10 GHz*2		A	—	4.6～4.8	—
Dissipation Factor	10 GHz*2		A	—	0.008～0.010	—
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14} ～1×10^{16}	2.5.17
Surface Resistance				Ω	1×10^{13} ～1×10^{15}	
Insulation Resistance			A	Ω	1×10^{14} ～1×10^{16}	—
			D-2/100		1×10^{12} ～1×10^{14}	—

*1)Heating Rate:10℃/min.(Comp.) *2)Measured by SPDR. *3)Refer to "Condition Note" *4)t0.8 mm

*Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thicness Code	Laminate thickness
MCL-E-700G	(R)	2, 3, 12 μm (LP, PF)	U0.04	0.04 mm
			U0.05	0.05 mm
			T0.06	0.06 mm
	(R)	2, 3, 12, 18 μm (LP) 2, 3, 12 μm (PF) 12, 18 μm (STD)	M0.06	0.06 mm
			0.1	0.11 mm
			M0.11	0.10 mm
			M0.15	0.16 mm
			M0.22	0.21 mm
			0.2	0.20 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.51	0.50 mm
			0.61	0.60 mm
			0.71	0.70 mm

Note1) STD: Standard copper foil, LP: Low profile copper foil, PF: Profile-free copper foil.
Note2) "U" for 1-ply; "T" for 2ply.
Note3) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type			Glass Cloth Style	Properties Resin Content (%)	Properties Dielectric Thickness after Lamination*1
GEA-700G	(—)	0.025	(1017N74)	1017	74±2	0.025 mm
		0.03	(1027N74)	1027	74±2	0.040 mm
		0.04	(1037N74)	1037	74±2	0.048 mm
		0.06	(1078N66)	1078	66±2	0.071 mm
		0.1	(2116N59)	2116	59±2	0.127 mm

Reference (IPC-TM-650)	2.3.16	—

*1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Halogen Free, High Tg, High Elastic Modulus, Low CTE Multilayer Material

MCL-E-770G （Type R)

Product

- **CCL MCL-E-770G(Type R)**
- **Prepreg GEA-770G**

Applications

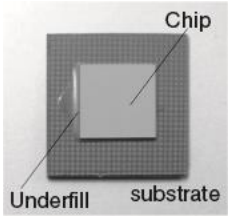
- **Semiconductor packages（FC-CSP, PoP, SiP）**
- **Thinner module**

Features

- MCL-E-770G has low CTE values in X,Y directions and reduces warpage of package substrate significantly.
- MCL-E-770G（Type RLH）has lower CTE value（≦ 2.0 ppm/℃）.

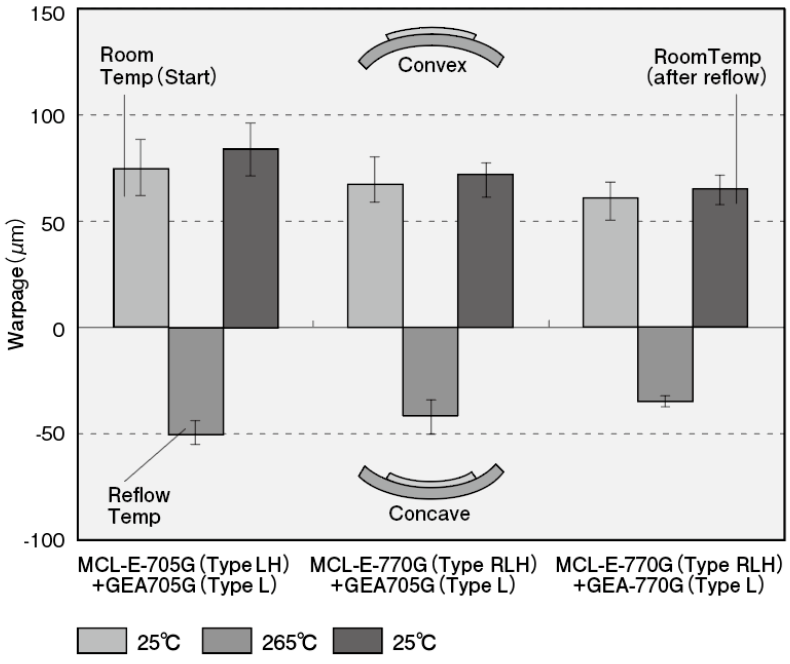
Characteristics

- **Warpage of FC-CSP**



TEG Chip
- ＊Chip size : 7.3 mm×7.3 mm
- ＊Chip thickness : 150 μm
- ＊Underfill thickness : 60 μm（CEL-C-3730-4）
- ＊Solder Resist thickness : 20 μm（ FZ-2700GA）

TEG Substrate
- ＊Package size: 14 mm×14 mm
- ＊Core Thickness 200 μm+1024(S-HD) prepreg



●Copper Clad Laminate

(t0.2 mm)

Item		Condition*3	Unit	Actual Value		Reference (IPC-TM-650)
				MCL-E-770G (Type R)	MCL-E-770G (Type RLH)	
Tg	TMA method	A	℃	260～280		2.4.24
	DMA method	A		300～330		—
CTE*1	X (30～120℃)	A	ppm/℃	4.0～6.0	1.5～2.0	—
	Y (30～120℃)			4.0～6.0	1.5～2.0	
Solder Heat Resistance (260℃)		A	sec.	≧ 300		—
T260 (Without copper)		A	min.	≧ 60		2.4.24.1
T288 (Without copper)				≧ 60		
Decomposition Temperature (TGA method 5% Weight Loss)		A	℃	430～450		2.3.40
Heat Resistance for HDI Proces (Semi-Additive)		260℃ Reflow	cycles	≧ 20		—
Copper Peel Strength	12μm	A	kN/m	0.7～0.9		2.4.8
	18μm			0.8～1.0		
Surface Roughness(Ra)		A	μm	2～3		2.2.17
Flexural Modulus (Lengthwise) *4		A	GPa	30～32	34～36	—
Dielectric Constant	10 GHz*2	A	—	4.2～4.4	3.9～4.1	—
Dissipation Factor	10 GHz*2	A	—	0.006～0.008	0.006～0.008	—
Volume Resistivity		C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}		2.5.17
Surface Resistance		C-96/40/90	Ω	1×10^{13}～1×10^{15}		
Insulation Resistance		A	Ω	1×10^{14}～1×10^{16}		—
		D-2/100		1×10^{12}～1×10^{14}		—

*1)Heating Rate:10℃/min. (Comp.) *2)Measured by SPDR. *3)Refer to "Condition Note" *4)t0.8 mm
*t0.4 mm thickness core is used depending on test item.
*Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-E-770G	(R)	2, 3, 12 μm (LP, PF)	T0.05	0.05 mm
			U0.04	0.04 mm
		2, 3, 12 , 18 μm (LP,PF) 12, 18 μm (STD)	M0.06	0.06 mm
			M0.11	0.10 mm
			0.2	0.20 mm
	(RLH)	2, 3, 12, 18 μm (LP) 2, 3, 12 μm (PF) 12, 18 μm (STD)	U0.04	0.04 mm
			M0.06	0.07 mm
			0.2	0.21 mm

Note1) STD: Standard copper foil, LP: Low profile copper foil, PF: Profile free copper foil.
Note2) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type		Glass Cloth Style	Properties: Resin Content (%)	Properties: Dielectric Thickness after Lamination*1
GEA-770G	—	≦0.025	(See P.18-19)		
		0.025 (1017N72)	1017	72±2	0.025 mm
		0.025 (1017N76)	1017	76±2	0.030 mm
		0.03 (1027N72)	1027	72±2	0.040 mm
		0.03 (1027N76)	1027	76±2	0.048 mm
		0.04 (1037N72)	1037	72±2	0.048 mm
	(L)	0.025 (L1017N72)	1017	72±2	0.025 mm
		0.025 (L1017N76)	1017	76±2	0.030 mm
		0.03 (L1027N72)	1027	72±2	0.040 mm
		0.03 (L1027N76)	1027	76±2	0.048 mm
		0.035 (L1024N68)	1024	68±2	0.041 mm
		0.035 (L1024N73)	1024	73±2	0.050 mm
		0.04 (L1037N72)	1037	72±2	0.048 mm
		0.045 (L1030N71)	1030	71±2	0.058 mm

Reference (IPC-TM-650)	2.3.16	—

※1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Thin Prepreg with Superior Surface Smoothness

GEA-705G, GEA-770G, GEA-795G

Product

- **Prepreg GEA-705G, GEA-770G , GEA-795G**

Applications

- **Semiconductor packages（FC-BGA, FC-CSP, PoP, SiP）**
- **HDI, PWB**
- **Thinner module, PWB**

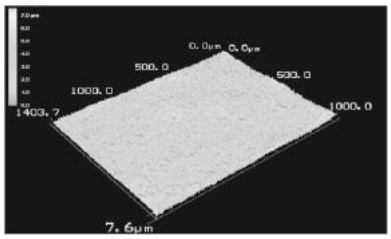
Features

- Superior in impedance control as prepreg thickness variation is small after press.
- Suitable for fine line formation with its smooth surface.
- Can achieve insulation layer thickness 16 μm.
- Suitable for coreless structure by superior variation of dimensional change.
- Undulation of TYPE-F prepreg is small after press. TYPE-F technology can reduce warpage at assembly.

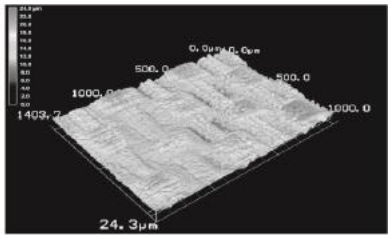
Characteristics

- **Surface Smoothness of Prepreg (Microscope)**

TYPE-F

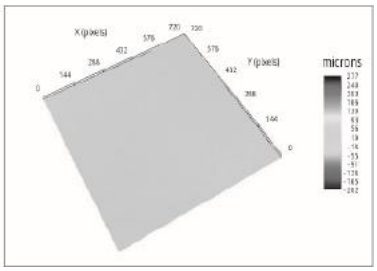


Conventional

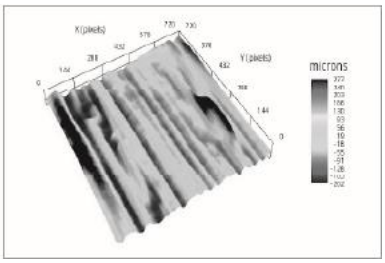


- **Panel Warpage of Prepreg after press (w/o copper, by shadow moire)**

TYPE-F

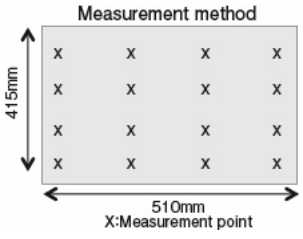


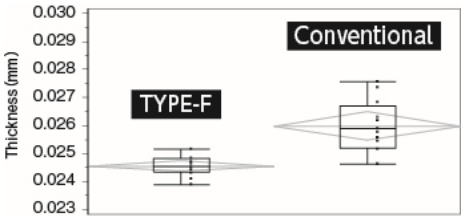
Conventional



- **Prepreg thickness after Press**

Laminate condition
Sample
・GEA-705G 1010F78
・Copper foil: 3 μm
Temp.: 230℃100 min
Heating rate: 3.0℃/min
Pressure: 3 MPa



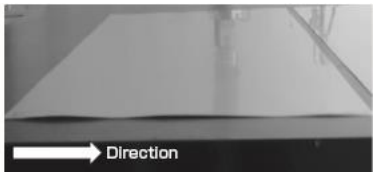


- **Undulation of prepreg after press(GEA-705G 1010F78)**

TYPE-F



Conventional



● Prepreg

(t0.025 mm)

Item		Condition *2	Unit	Actual Value			Test Method (IPC-TM-650)
				GEA-705G	GEA-770G	GEA-795G	
Tg	TMA method	A	℃	250～270	260～280	260～290	2.4.24.5
	DMA method	A		295～305	200～300	315～345	—
CTE *1	X（30～120℃）	A	ppm/ ℃	8～10	6～8	5.5～7.5	2.4.24.5
	Y（30～120℃）			8～10	6～8	5.5～7.5	
T-260（Without Copper）		A	min.	≧60	≧60	≧60	2.4.24.1
T-288（Without Copper）				≧60	≧60	≧60	
Undulation（After Press）		A	mm	0.0～0.3	0.0～0.3	0.0～0.3	—

*1) Heating Rate: 10℃/min. (Tensile)　*2)Refer to"Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

● Prepreg

Part Number	Type		Glass Cloth Style	Properties Resin Content (%)	Properties Dielectric Thickness after laminate *1
GEA-705G	0.015	(1006F70)	1006	70±2	0.016 mm
		(1006F72)		72±2	0.017 mm
		(1006F74)		74±2	0.018 mm
		(1006F76)		76±2	0.020 mm
		(1006F78)		78±2	0.022 mm
	0.020	(1010F74)	1010	74±2	0.021 mm
		(1010F76)		76±2	0.023 mm
		(1010F78)		78±2	0.025 mm
		(1010F80)		80±2	0.027 mm
		(1010F84)		84±2	0.035 mm
	0.025	(1017F73)	1017	73±2	0.025 mm
		(1017F78)		78±2	0.031 mm
		(1017F83)		83±2	0.040 mm
GEA-770G	0.015	(1006F74)	1006	74±2	0.017 mm
		(1006F76)		76±2	0.019 mm
		(1006F78)		78±2	0.021 mm
		(1006F80)		80±2	0.023 mm
	0.020	(1010F67)	1010	67±2	0.016 mm
		(1010F69)		69±2	0.018 mm
		(1010F71)		71±2	0.019 mm
		(1010F73)		73±2	0.021 mm
		(1010F78)		78±2	0.026 mm
		(1010F82)		82±2	0.032 mm
	0.025	(1017F73)	1017	73±2	0.026 mm
		(1017F78)		78±2	0.033 mm
		(1017F83)		83±2	0.043 mm
	0.030	(1027F73)	1027	73±2	0.042 mm
		(1027F78)		78±2	0.052 mm
GEA-795G	0.020	(1010F72)	1010	72±2	0.019 mm
		(1010F74)		74±2	0.021 mm
		(1010F76)		76±2	0.022 mm
		(1010F78)		78±2	0.025 mm
		(1010F80)		80±2	0.027 mm
		(1010F82)		82±2	0.030 mm
	0.025	(1017F72)	1017	72±2	0.023 mm
		(1017F74)		74±2	0.025 mm
		(1017F76)		76±2	0.027 mm
		(1017F78)		78±2	0.030 mm
		(1017F80)		80±2	0.033 mm
		(1017F82)		82±2	0.037 mm

Reference（IPC-TM-650）	2.3.16	—

*1）The dielctric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Value change depending on the press condition or inner layer pattern.

Halogen Free, High Elastic Modulus, Low CTE Multilayer Material

MCL-E-679FG(Type R)/(Type S)

Product

- **CCL MCL-E-679FG(Type R)/(Type S)**
- **Prepreg GEA-679FG**

MCL-E-679FGB(Type R)/(Type S)<Black Type>

Applications

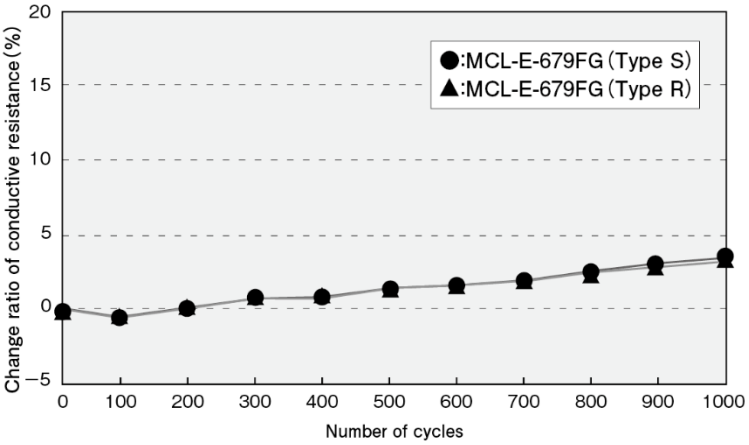
- **Semiconductor packages（FC-BGA, BGA, CSP）**
- **Build up PWB**
- **Personal computer, high density electronic equipment**

Features

- Halogen free material for environmental concerns. ●CTE（Z-direction）is 50% lower than that of our standard FR-4.
- Elastic modulus is 20% higher than that of our standard FR-4. Even thin laminate has less warpage and deflection.
- Superior heat resistance for soldering（suitable for lead free process）.
- Suitable for fine patterning with 1/4 surface roughness of our standard FR-4 pattern possible.

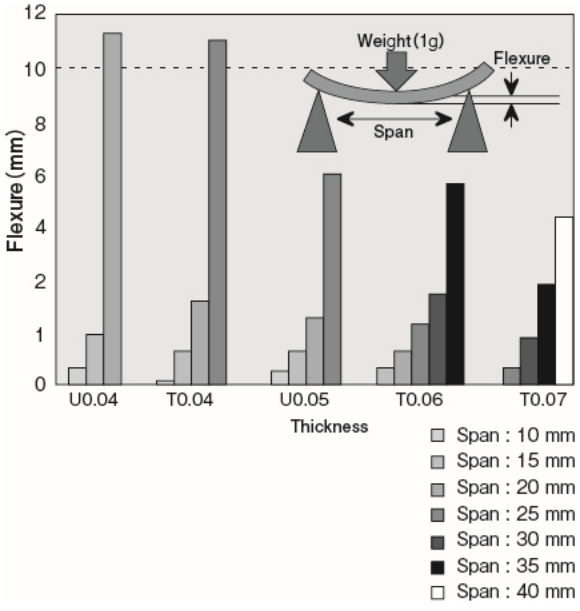
Characteristics

●Through-hole reliability

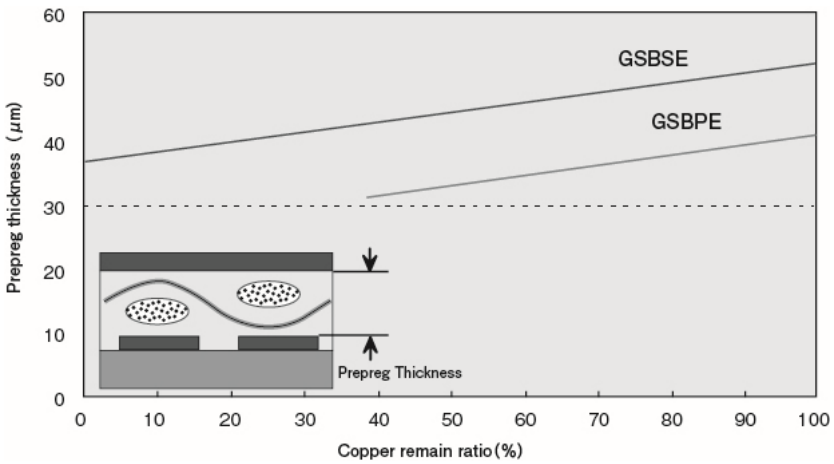


Test condition: −55℃, 30 min. ⇔150℃, 30 min.
Pattern: Wall to wall distance 0.3 mm, Laminate thickness: t0.8 mm
Pre-condition: 260℃ reflow × 2 times⇒Solder dipping（260℃ 10 sec.）

●Stiffness Properties



●Prepreg thickness after pattern filling （Inner layer copper 15μm）



MCL-E-679FGB(Type R)/(Type S)<Black Type>

●Copper Clad Laminate

（t0.4 mm, t0.8 mm）

Item			Condition*3	Unit	Actual Value				Reference（IPC-TM-650）
					MCL-E-679FGB（Type R）	MCL-E-679FG（Type R）	MCL-E-679FGB（Type S）	MCL-E-679FG（Type S）	
Tg	TMA method		A	℃	165～175		175～185		2.4.24
	DMA method		A		200～220		210～230		—
CTE*1	X（30～120℃）		A	ppm/℃	13～15		12～14		—
	Y（30～120℃）				13～15		12～14		
	Z	（< Tg）	A		23～33		20～30		2.4.24
		（> Tg）	A		140～170		130～160		
Solder Heat Resistance（260℃）			A	sec.	≧300				—
T-260（Without Copper）			A	min.	≧60				2.4.24.1
T-288（Without Copper）					≧60				
Decomposition Temperature（TGA method 5% Weight Loss）			A	℃	340～360				2.3.40
Heat Resistance for HDI Process			260℃ Reflow	cycles	≧10				—
Copper Peel Strength	18μm		A	kN/m	0.9～1.1		1.1～1.2		2.4.8
	35μm				1.1～1.2		1.2～1.3		
Surface Roughness（Ra）			A	μm	2～3				2.2.17
Flexural Modulus（Lengthwise）			A	GPa	23～28		24～29		2.4.4
Dielectric Constant	1 GHz*2		A	—	4.9～5.2				2.5.5.9
	10 GHz*2				4.8～5.1				JPCA TM-001
Dissipation Factor	1 GHz*2		A	—	0.0130～0.0150				2.5.5.9
	10 GHz*2				0.0140～0.0160				JPCA TM-001
Volume Resistivity			C-96/40/90	Ω･cm	1×1014～1×1016				2.5.17
Surface Resistance				Ω	1×1013～1×1015				
Insulation Resistance			A	Ω	1×1014～1×1016				—
			D-2/100		1×1012～1×1014				—

*1)Heating Rate:10℃/min.(Comp.)　*2)Measured by SPDR method.　*3)Refer to "Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copperfoil Thickness	Thickness Code	Laminate Thickness
MCL-E-679FG MCL-E-679FGB	（S）	3,12,18 μm (LP) 12,18 μm (STD)	T0.07	0.07 mm
			M0.06	0.07 mm
	（R） （S）	3, 12, 18 μm (LP) 12, 18, 35 μm (STD)	0.1	0.11 mm
			0.2	0.20 mm
			0.41	0.40 mm
			0.81	0.80 mm

Note1) STD:Standard copper foil, LP:Low profile copper foil.
Note2) "U" for 1-ply; "T" for 2 ply.
Note3) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type			Glass Cloth	Properties	
				Style	Resin Content（%）	Dielectric Thickness after Lamination*1
GEA-679FG*2	（R）	0.03	（GBPE）	1027	73±2	0.040 mm
		0.04	（GRZPE）	1037	73±2	0.048 mm
		0.06	（GRROE）	1078	68±2	0.079 mm
		0.1	（GRSKE）	2116	58±2	0.127 mm
	（S）	0.03	（GSBPE）	1027	73±2	0.040 mm
		0.03	（GSBSE）	1027	78±2	0.050 mm
		0.04	（GSZPE）	1037	73±2	0.048 mm
		0.06	（GSROE）	1078	68±2	0.079 mm
		0.1	（GSSKE）	2116	58±2	0.127 mm

Reference（IPC-TM-650）	2.3.16	—

*1)The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.
*2)Black type is not available for prepreg.

Halogen Free, High Tg, Low Transmission Loss, Low CTE Multilayer Material

MCL-HS200

Product

- **CCL MCL-HS200**
- **Prepreg GH-200**

Applications

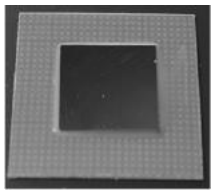
- **Semiconductor packages (FC-CSP, PoP, SiP)**
- **Thinner Module PWB**

Features

- MCL-HS200 has hybrid properties of package and high speed materials.
- MCL-HS200 has low CTE value in X,Y directions and good dielectric property (Dk 3.7,Df 0.0040 at 10 GHz).
- MCL-HS200(Type D) achieved low dielectric constant (3.4 at 10 GHz) and low dissipation factor (0.0025 at 10 GHz) using low Dk glass
- Well-suited for build-up construction.

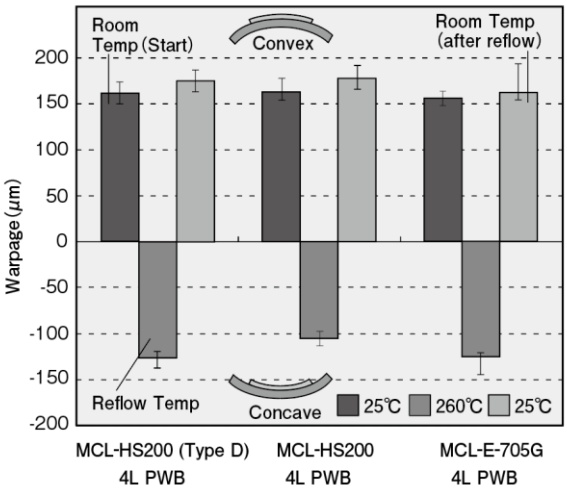
Characteristics

- **Warpage of 4 layer board**

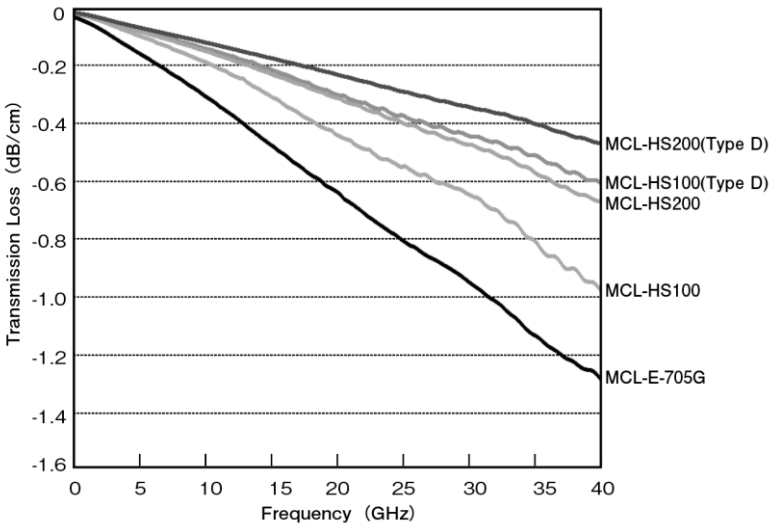


TEG Chip
- Chip size 10 mm x 10 mm
- Chip thickness 100 μm

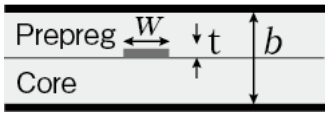
TEG Substrate
- Package size 14 mm x 14 mm
- L1,4: 12 μm Cu: 65%, L2,3 No Copper SR: 18 μ
- Core thickness : 45 μm
- Prepreg thickness : 45 μm



- **Transmission Loss**



<Measurement conditions>
*Evaluation PWB : Strip-line
*Temperture and Humidity : 25℃/60%RH
*Characteristic impedance : Approx. 50 Ω
*Proofreading method : TRL(Thru-Reflect-Line)



*Trace width (w) : 0.12～0.14 mm
*Dielectric thickness(b) : 0.65 mm
*Trace thickness(t) : 18 μm

●Copper Clad Laminate

(t0.2 mm, t0.4 mm)

Item			Condition*3	Unit	Actual Value		Reference (IPC-TM-650)
					MCL-HS200	MCL-HS200 (Type D)	
Tg	TMA method		A	℃	250～270		2.2.24
	DMA method		A		330～350		—
CTE*1	X*2(30～120℃)		A	ppm/℃	8～10		—
	Y*2(30～120℃)		A		8～10		
	Z	(<Tg)	A		25～35		2.2.24
		(>Tg)			140～200		
Solder Heat Resistance (260℃)			A	sec.	≧300		—
T260 (Without copper)			A	min.	≧60		2.4.24.1
T288 (Without copper)			A		≧60		
Decomposition Temperature (TGA method 5% weight loss)			A	℃	410～430		2.3.40
Copper Peel strength	12 μm		A	kN/m	0.5～0.7		2.2.8
	18 μm		A		0.6～0.8		
Surface roughness (Ra)			A	μm	2～3		2.2.17
Flexual Modulus (Lengthwise)			A	GPa	21～26		2.4.4
Dielectric Constant*2	10 GHz*2		A	—	3.6～3.8	3.2～3.4	IEC-62810
Dissipation Factor*2	10 GHz*2		A	—	0.0035～0.0045	0.0020～0.0030	
Volume Resistivity			C-96/20/65+C-96/40/90	Ω·cm	1×10^{14} ～1×10^{16}		2.5.17
Surface resistance			C-96/20/65+C-96/40/90	Ω	1×10^{13} ～1×10^{15}		
Insulation resistance			C-96/20/65	Ω	1×10^{14} ～1×10^{16}		—
			C-96/20/65+D-2/100		1×10^{13} ～1×10^{15}		—

*1) Heating rate : 10℃/min.(Comp.) *2) Measured by cavity resonator. *3) Refer to "Condition Note" *Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-HS200	—	1.5, 3, 12 μm (LP) 18 μm (RT) 12, 18 μm (HVLP)	U0.04	0.04 mm
		1.5, 3, 12 μm (LP) 18, 35 μm (RT) 12, 18, 35 μm (HVLP)	M0.06	0.06 mm
			0.1	0.10 mm
			0.2	0.20 mm
			0.41	0.41 mm
	(D)	1.5, 3, 12 μm (LP) 18 μm (RT) 12, 18 μm (HVLP)	U0.04	0.04 mm
		1.5, 3, 12 μm (LP) 18, 35 μm (RT) 12, 18, 35 μm (HVLP)	M0.06	0.06 mm
			0.1	0.10 mm
			0.2	0.20 mm
			0.41	0.41 mm

Note1) LP/HVLP: Low profile copper foil. Note2) Laminate means dielectric layer thickness.

●Prepreg

Part Number	Type			Glass cloth	Properties	
				Style	Resin content (%)	Dielectric Thickness after Lamination*1
GH-200	(D)	0.025	(See P.26-27)			
	—	0.05	(1037N73)	1037	73±2	0.051 mm
		0.05	(1037N77)	1037	77±2	0.061 mm
		0.06	(1078N63)	1078	63±2	0.072 mm
		0.06	(1078N73)	1078	67:±2	0.082 mm
		0.08	(3313N60)	3313	60±2	0.110 mm
		0.08	(3313N63)	3313	63±2	0.121 mm
	(D)	0.05	(D1035N69)	1037	69±2	0.051 mm
		0.05	(D1035N73)	1037	73±2	0.059 mm
		0.06	(D1035N75)	1035	75±2	0.064 mm
		0.06	(D1078N65)	1078	65±2	0.072 mm
		0.06	(D1078N69)	1078	69±2	0.083 mm
		0.08	(D3313N62)	3313	62±2	0.108 mm
		0.08	(D3313N65)	3313	65±2	0.118 mm

Reference (IPC-TM-650)					2.3.16	—

*1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Halogen Free, High Tg, Low Transmission Loss, Low CTE Multilayer Material

MCL-HS100

Product

- **CCL MCL-HS100**
- **Prepreg GH-100**

Applications

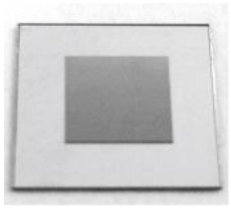
- **Semiconductor packages（FC-CSP, PoP, SiP）**
- **Thinner Module PWB**

Features

- MCL-HS100 has hybrid properties of package and high speed materials.
- MCL-HS100 has low CTE value in X,Y directions and good dielectric property.（Dk 3.7,Df 0.0040 at 10 GHz）
- MCL-HS100（Type D）achieved low dielectric constant（3.4 at 10 GHz ）
  and low dissipation factor（0.0025 at 10 GHz）using low Dk glass.
- Well-suited for build-up construction.

Characteristics

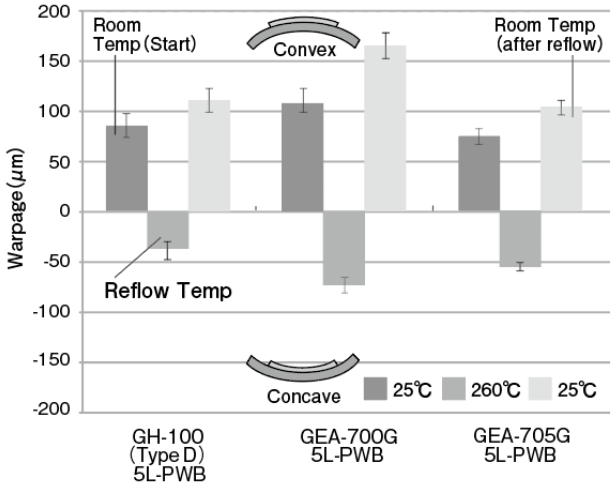
- **Warpage of coreless-5layer**



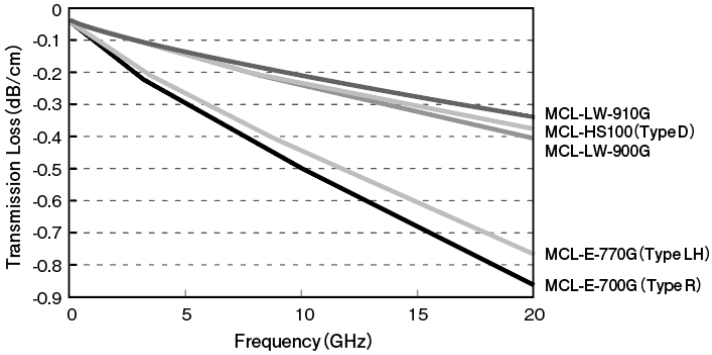
TEG Chip
- ＊Package size : 14 mm×14 mm
- ＊Chip size : 7.3 mm×7.3 mm
- ＊Chip thickness : 150 μm
- ＊Underfill thickness : 60 μm（CEL-C-3730-4）

TEG Substrate
- ＊L1, 5:12 μm Cu100%, L2, 3, 4: No copper,SR: -

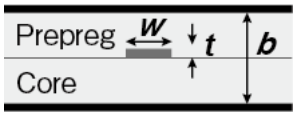
Prepreg construction
- ＊GH-100（TypeD）（1078, R.C.: 63%）×4 ply
- ＊GEA-700G（1078, R.C.: 66%）×4 ply
- ＊GEA-705G（1078, R.C.: 65%）×4 ply



- **Transmission Loss**



《Measurement conditions》
- Evaluation PWB : Strip-line
- Temperature & Humidity:25℃/60%RH
- Characteristic impedance : Approx.50Ω
- Proofreading method : TRL（Thru-Reflect-Line）



- Trace width（w） : 0.12～0.14 mm
- Dielectric thicknes（b） : 0.25 mm
- Trace thicknes（st） : 18 μm

● Copper Clad Laminate

(t0.2 mm, t0.4 mm)

Item			Condition*3	Unit	Actual Value		Reference (IPC-TM-650)
					MCL-HS100	MCL-HS100 (Type D)	
Tg	TMA method		A	℃	240 ～ 260		2.4.24
	DMA method		A		240 ～ 260		—
CTE*1	X (30 ～ 120℃)		A	ppm/℃	6 ～ 8		—
	Y (30 ～ 120℃)		A		6 ～ 8		
	Z	(＜Tg)	A		20 ～ 30		2.4.24
		(＞Tg)	A		130 ～ 180		
Solder Heat Resistance (260℃)			A	sec.	≧300		—
T-260 (Without copper)			A	min.	≧60		2.4.24.1
T-288 (Without copper)					≧60		
Decomposition Temperature (TGA method 5% weight loss)			A	℃	430 ～ 450		2.3.40
Copper Peel Strength	12 μm		A	kN/m	0.5 ～ 0.7		2.4.8
	18 μm				0.5 ～ 0.7		
Surface Roughness (Ra)			A	μm	2 ～ 3		2.2.17
Flexural Modulus (Lengthwise)			A	GPa	23 ～ 28		2.4.4
Dielectric Constant	10 GHz*2		A	—	4.0 ～ 4.2	3.3 ～ 3.5	IEC-62810
Dissipation Factor	10 GHz*2		A	—	0.0050 ～ 0.0060	0.0025 ～ 0.0035	
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}		2.5.17
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}		
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}		—
			D-2/100		1×10^{13}～1×10^{15}		—

*1) Heating Rate: 10℃/min.(Comp.) *2) Measured by Cavity Resonator. *3) Refer to "Condition Note"

※Above data are experimental results and not guaranteed.

Standard Specifications

● Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-HS100	—	1.5, 3, 12 μm (LP) 18 μm (RT) 12, 18 μm (HVLP)	U0.04	0.04 mm
		1.5, 3,12 μm (LP) 18, 35 μm (RT) 12, 18, 35 μm (HVLP)	M0.06	0.06 mm
			0.1	0.10 mm
			0.2	0.20 mm
			0.41	0.40 mm
	(D)	1.5, 3,12 μm (LP) 18 μm (RT) 12, 18 μm (HVLP)	U0.04	0.04 mm
		1.5, 3,12 μm (LP) 18, 35 μm (RT) 12, 18, 35 μm (HVLP)	M0.06	0.06 mm
			0.1	0.10 mm
			0.2	0.20 mm
			0.41	0.40 mm

Note1) LP/HVLP: Low profile copper foil.

Note2) Laminate thickness means dielectric layer thickness.

● Prepreg

Part Number	Type			Glass cloth	Properties	
				Style	Resin Content(%)	Dielectric Thickness after Lamination※1
GH-100	(D)	0.025	(See P.26-27)			
	—	0.03	(1027N71)	1027	71±2	0.042 mm
		0.05	(1037N71)	1037	71±2	0.050 mm
		0.06	(1078N61)	1078	61±2	0.071 mm
		0.1	(2116N55)	2116	55±2	0.130 mm
	(D)	0.03	(D1027N73)	1027	73±2	0.042 mm
		0.05	(D1037N74)	1037	74±2	0.050 mm
		0.06	(D1078N63)	1078	63±2	0.072 mm
		0.1	(D2116N57)	2116	57±2	0.130 mm
Reference (IPC-TM-650)					2.3.16	—

※1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Low Transmission Loss, Thin Prepreg with Superior Surface Smoothness

GH-100(Type D)、GH-200(Type D)

Product

- **Prepreg　GH-100(Type D)、GH-200(Type D)**　TYPE-F

Applications

- **Semiconductor packages（FC-CSP, PoP, SiP)**
- **Thinner Module PWB**

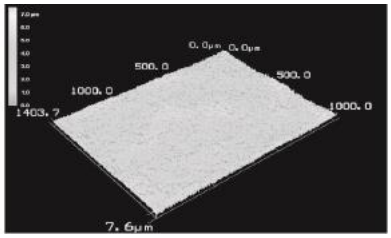
Features

- GH-100 (Type D) and GH-200 (Type D) have hybrid properties of package and high speed materials.
- Good for impedance control as thickness variation is small after press.
- Halogen free material with low CTE values in X, Y directions (α1、α2).
- Good dielectric properties by applying low dielectric glass cloths.

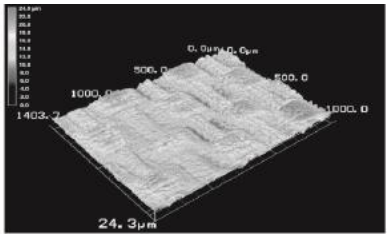
Characteristics

- **Surface Smoothness of Prepreg (Microscope)**

TYPE-F

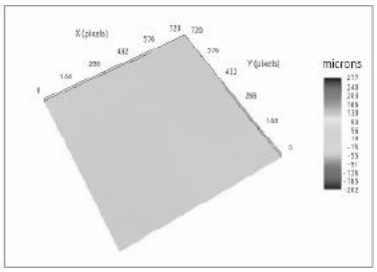


Conventional

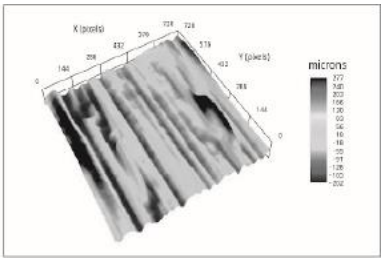


- **Panel Warpage of Prepreg after press (w/o copper, by shadow moire)**

TYPE-F

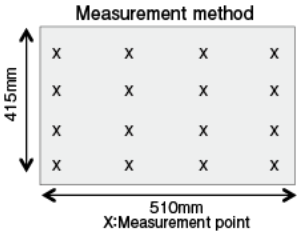


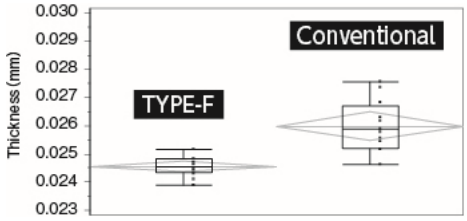
Conventional



- **Prepreg thickness after Press**

Laminate condition
Sample
・GEA-705G 1010F78
・Copper foil ; 3 μm
Temp. ; 230℃100 min
Heating rate ; 3.0℃/min
Pressure ; 3 MPa





- **Undulation of prepreg after press(GEA-705G 1010F78)**

TYPE-F



Conventional



Characteristics

● Prepreg

（t0.025 mm）

Item		Condition*3	Unit	Actual Value		Test Method (IPC-TM-650)
				GH-100(Type D)	GH-200(Type D)	
Tg	TMA method	A	℃	240～260	220～240	2.4.24.5
	DMA method	A		240～260	250～270	-
CTE*1	X(30～120℃)	A	ppm/℃	13～15	14～16	2.4.24.5
	Y(30～120℃)			13～15	14～16	
T260(Without Copper)		A	min.	≧60	≧60	2.4.24.1
T 288(Without Copper)				≧60	≧60	
Dielectric Constant*2	10 GHz	A	-	3.3～3.5	3.2～3.4	IEC-62610
Dissipation Factor*2	10 GHz	A	-	0.0030～0.0050	0.0020～0.0040	

*1) Heating Rate: 10℃/min. (Tensile)　*2)Measured by cavity resonator method.　*3)Refer to " Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

● Prepreg

Part Number	Type			Glass Cloth	Properties	
				Style	Resin Content (%)	Dielectric Thickness after lamination
GH-100	(D)	0.025	1017F71K	1017	71±2	0.027 mm
			1017F73K		73±2	0.029 mm
			1017F75K		75±2	0.031 mm
			1017F77K		77±2	0.034 mm
			1017F79K		79±2	0.038 mm
GH-200	(D)	0.020	D1010F72K	1010	72±2	0.019 mm
			D1010F74K		74±2	0.021 mm
			D1010F76K		76±2	0.023 mm
			D1010F78K		78±2	0.025 mm
			D1010F80K		80±2	0.028 mm
			D1010F82K		82±2	0.031 mm
			D1010F84K		84±2	0.035 mm
		0.025	1017F68K	1017	68±2	0.020 mm
			1017F70K		70±2	0.022 mm
			1017F72K		72±2	0.024 mm
			1017F74K		74±2	0.026 mm
			1017F76K		76±2	0.028 mm
			1017F78K		78±2	0.031 mm
			1017F80K		80±2	0.034 mm
			1017F82K		82±2	0.038 mm
			1017F84K		84±2	0.043 mm
		0.030	D1027F66K	1027	66±2	0.031 mm
			D1027F68K		68±2	0.033 mm
			D1027F70K		70±2	0.035 mm
			D1027F72K		72±2	0.038 mm
			D1027F74K		74±2	0.041 mm
			D1027F76K		76±2	0.045 mm
			D1027F78K		78±2	0.049 mm
Reference(IPC-TM-650)					2.3.16	-

*1) The dielctric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0 %.
Values change depending on the press condition or inner layer pattern.

High Dk Multilayer Material

MCL-HD60/HD80/HD90

Product

- **CCL MCL-HD60/HD80/HD90**
- **Prepreg GD-60/GD-80/GD-90**

Applications

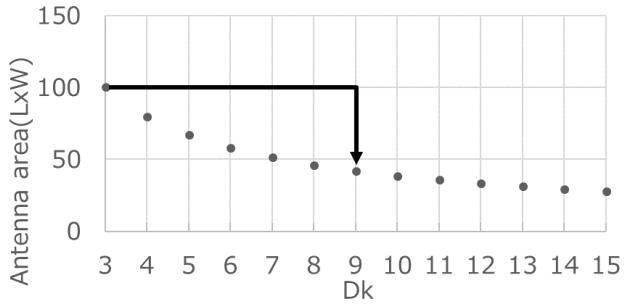
- **Antenna-in-package**

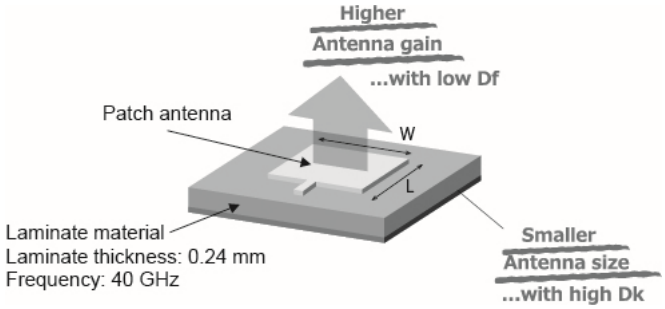
Features

- Enables miniaturization and higher anntena gain of antenna-in-package.
- High Dk and low Df substrate material.
- High Dk with low transmission loss.
- Excellent reliability (heat resistance, insulation, connection)

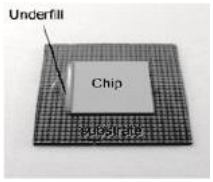
Characteristics

- **Antenna size simulation**



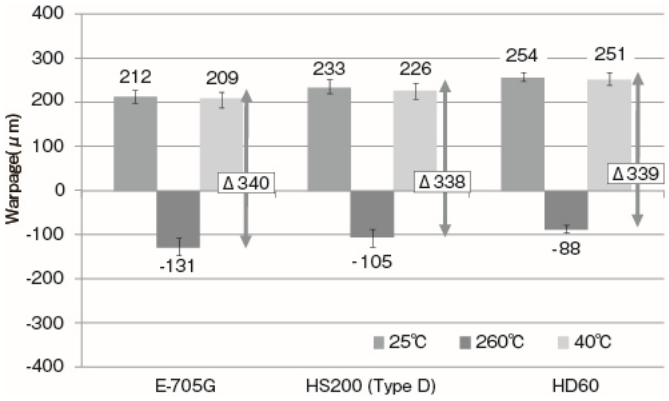


- **Mounting warpage characteristic evaluation results**



Sample Construction
(Substrate : 14mm × 14 mm)
- E-705G(1037, 60 μm) x 3 ply
- HS200(Type D)(1037, 60 μm) x 3 pl
- HD60 (1037, 60 μm) x 3 ply
- Copper foil : 3 μm, 65%
- Solder resist : 20 μm (SR-FA)

TEG Chip
- Chip size : 10 mm×10 mm
- Chip thickness : 100 μm
- Chip bump : N/A (Gap:60 μm)
- Underfill : CEL-C-3730 series



- **Heat resistance test results (reflow resistance test)**

Test sample : 4L construction, 35 mm x 35 mm
- L1, L4 copper thickness: 1.5 μm
- L2, L3 copper thickness: 18 μm Cu Coverage : 15-90% surface treatmen : t CZ8401

Test Condition

Conditon1 ●Pre-condition : 144 hrs 60℃/60% RH ●Test condition : 260℃ reflow x 30 times

Conditon2 ●Pre-condition : PCT-4 hrs ●Test condition : 260℃ reflow x 5 times

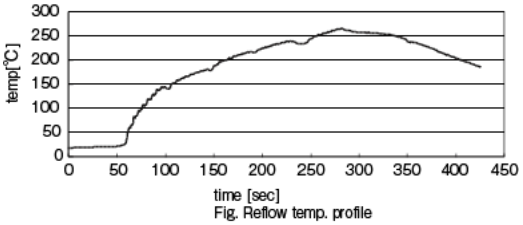


Fig. Reflow temp. profile

Item		#1	#2	#3	#4
Construction	Core	HS200(Type D)	HD60	HS200(Type D)	HD60
Reflow resistance	60℃/60%RH 144h	30 pass	30 pass	30 pass	30 pass
	PCT-4h	5 pass	5 pass	5 pass	5 pass

●Copper Clad Laminates

Item		Unit	Actual Value					Reference Standards (IPC-TM-650)
			HD60	HD60HD90(Type D)	HD80	HD90	HD90HD90(Type D)	
Glass cloth	Glass type	-	E	Low Dk	E	E	Low Dk	—
	Style	-	2116	2116	2116	2116	2116	—
Tg	DMA	℃	230 - 240	230 - 240	230 - 240	230 - 240	230 - 240	—
CTE	XY*1	ppm/℃	8 - 10	8 - 10	9 - 11	9 - 11	9 - 11	—
Copper Peel Strength	12 μm (LP)	kN/m	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	0.5 - 0.6	2.4.8
Elastic Modulus		GPa	19 - 23	18 - 22	18 - 22	18 - 22	17 - 21	—
Dielectric Constant	10 GHz*2	—	6.0 - 6.2	5.8 - 6.0	8.0 – 8.2	9.0 - 9.2	8.2 - 8.4	—
	20 GHz*2	—	6.1 - 6.3	5.9 - 6.1	8.0 – 8.2	9.0 - 9.2	8.2 - 8.4	—
	36 GHz*3	—	6.1 - 6.3	5.9 - 6.1	8.0 – 8.2	9.0 - 9.2	8.2 - 8.4	—
	50 GHz*3	—	6.1 - 6.3	5.9 - 6.1	8.0 – 8.2	9.0 - 9.2	8.2 - 8.4	—
	60 GHz*3	—	6.1 - 6.3	5.9 - 6.1	8.0 – 8.2	9.0 - 9.2	8.2 - 8.4	—
Dissipation Factor	10 GHz*2	—	0.0050 - 0.0060	0.0040 - 0.0050	0.0050 - 0.0060	0.0055 - 0.0065	0.0045 - 0.0055	—
	20 GHz*2	—	0.0055 - 0.0065	0.0045 - 0.0055	0.0055 - 0.0065	0.0060 - 0.0070	0.0050 - 0.0060	—
	36 GHz*3	—	0.0060 - 0.0070	0.0050 - 0.0060	0.0060 - 0.0070	0.0065 - 0.0075	0.0055 - 0.0065	—
	50 GHz*3	—	0.0065 - 0.0075	0.0055 - 0.0065	0.0065 - 0.0075	0.0070 - 0.0080	0.0060 - 0.0070	—
	60 GHz*3	—	0.0065 - 0.0075	0.0055 - 0.0065	0.0065 - 0.0075	0.0070 - 0.0080	0.0060 - 0.0070	—

*1) Heating Rate: 10℃/min. (Tensile) *2) SPDR method, Drying:105℃ / 1h. *3) Cut-off circular waveguide method, Drying:105 ℃ / 1h.
※Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-HD60	—	1.5, 2, 3, 12, 18 μm (VLP) 12, 18 μm (HVLP)	0.11	0.10 mm
			0.21	0.20 mm
			0.24	0.24 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.51	0.50 mm
	(D)	1.5, 2, 3, 12, 18 μm (VLP) 12, 18 μm (HVLP)	0.11	0.10 mm
			0.21	0.20 mm
			0.24	0.24 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.51	0.50 mm
MCL-HD80	—	1.5, 2, 3, 12, 18 μm (VLP) 12, 18 μm (HVLP)	0.11	0.10 mm
			0.21	0.20 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.81	0.80 mm
			1.21	1.20 mm
			1.61	1.60 mm
MCL-HD90	—	1.5, 2, 3, 12, 18 μm (VLP) 12, 18 μm (HVLP)	0.11	0.10 mm
			0.21	0.20 mm
			0.24	0.24 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.81	0.80 mm
			1.21	1.20 mm
			1.61	1.60 mm
	(D)	1.5, 2, 3, 12, 18 μm (VLP) 12, 18 μm (HVLP)	0.11	0.10 mm
			0.21	0.20 mm
			0.24	0.24 mm
			0.31	0.30 mm
			0.41	0.40 mm
			0.51	0.50 mm

Note1) LP/HVLP: Low profile copper foil.
Note2) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type			Glass Cloth	Properties	
				Style	Resin Content(%)	Dielectric Thickness after Lamination*1
GD-60	—	0.025	(1017F80)	1017	80±2	0.031 mm
			(1017F88)		88±2	0.053 mm
		0.030	(1027F74)	1027	74±2	0.038 mm
			(1027F78)		78±2	0.045 mm
		0.040	(1037F79)	1037	79±2	0.056 mm
			(1037F82)		82±2	0.065 mm
		0.060	(1078F67)	1078	67±2	0.070 mm
			(1078F73)		73±2	0.087 mm
			(1078F76)		76±2	0.098 mm
	(D)	0.025	(D1017F82)	1017	82±2	0.033 mm
			(D1017F90)		90±2	0.060 mm
		0.030	(D1027F76)	1027	76±2	0.038 mm
			(D1027F80)		80±2	0.046 mm
		0.040	(D1037F81)	1037	81±2	0.057 mm
			(D1037F84)		84±2	0.068 mm
		0.060	(D1078F69)	1078	69±2	0.069 mm
			(D1078F75)		72±2	0.077 mm
			(D1078F78)		78±2	0.099 mm
GD-80	—	0.025	(1017F82)	1017	82±2	0.034 mm
			(1017F90)		90±2	0.063 mm
		0.030	(1027F74)	1027	74±2	0.037 mm
			(1027F78)		78±2	0.045 mm
		0.040	(1037F78)	1037	78±2	0.052 mm
			(1037F82)		82±2	0.064 mm
		0.060	(1078F68)	1078	68±2	0.071 mm
			(1078F72)		72±2	0.082 mm
			(1078F76)		76±2	0.097 mm
GD-90	—	0.025	(1017F82)	1017	82±2	0.034 mm
			(1017F90)		90±2	0.063 mm
		0.030	(1027F74)	1027	74±2	0.037 mm
			(1027F78)		78±2	0.044 mm
		0.040	(1037F78)	1037	78±2	0.052 mm
			(1037F82)		82±2	0.064 mm
		0.060	(1078F68)	1078	68±2	0.071 mm
			(1078F72)		72±2	0.081 mm
			(1078F76)		76±2	0.096 mm
	(D)	0.025	(D1017F84)	1017	84±2	0.036 mm
			(D1017F92)		92±2	0.073 mm
		0.030	(D1027F76)	1027	76±2	0.037 mm
			(D1027F80)		80±2	0.045 mm
		0.040	(D1037F80)	1037	80±2	0.052 mm
			(D1037F84)		84±2	0.066 mm
		0.060	(D1078F70)	1078	70±2	0.070 mm
			(D1078F74)		74±2	0.082 mm
			(D1078F78)		78±2	0.097 mm

*1)The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0 %.
Values change depending on the press condition or inner layer pattern.

Low Transmission Loss, Low CTE Adhesive Film

MCF-200HS

Product

- **Copper Clad Film MCF-200HS**

Applications

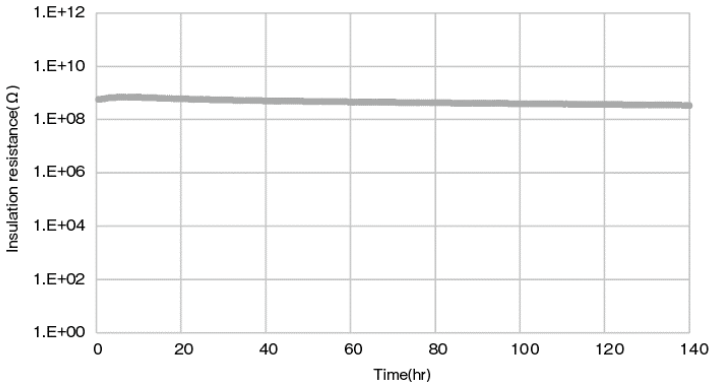
- **Semiconductor packages (FC-CSP, PoP, SiP)**
- **Thinner Module PWB**

Features

- Film material (copper clad film) that can achieve an insulation layer of 15 μm or less.
- Excellent dielectric properties, contributing to thinner insulation layer thickness and lower transmission loss
- Excellent insulation properties

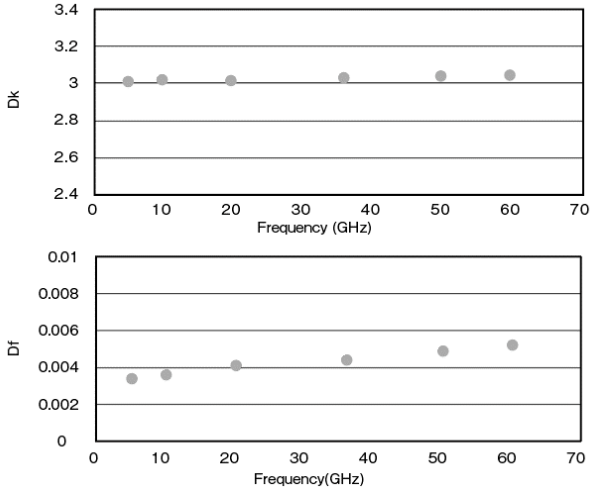
Characteristics

- **Reliability test (Layer to layer)**



・Sample: MCF-200HS Dielectric thickness : 20 μm
・Pre-treatment condition: JEDEC Level 2 (60 ℃/60 %RH 120hr + 260 ℃ reflow 6 cycle)
・Evaluation condition: 130 ℃/85 %RH, 6 V

- **Dielectric properties (Frequency drift)**



Sample MCF-200HS Dielectric thickness：20μm
Evaluation method 5～20GHz：SPDR
36～60GHz：Circular Cut-off Waveguide method

- **Copper Clad Film**

Item		Condition*3	Unit	Actual Value MCF-200HS	Reference (IPC-TM-650)
Tg	DMA	A	℃	250～270	—
CTE*1	X(30～120℃)	A	ppm/℃	35～40	—
	Y(30～120℃)			35～40	
Solder Heat Resistance(260℃)		A	sec	≧300	—
Copper Peel Strength	3 μm	A	kN/m	0.5～0.7	2.4.8
	12 μm (LP)			0.5～0.7	
Elastic Modulus (Tensile)		A	GPa	5～7	—
Dielectric Constant	10 GHz*2	A	—	2.9～3.1	—
Dissipation Factor	10 GHz*2	A	—	0.003～0.004	—

*1) Heating Rate: 10℃/min. (Tensile) *2) Measured by Split post dielectric resonators (SPDR). *3) Refer to "Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

- **Copper Clad Film**

Part Number	Copper foil type	Dielectric thickness
MCF-200HS	1.5, 3, 12 μm (LP)	15 ± 2 μm
		20 ± 2 μm
		25 ± 2 μm

Low CTE Adhesive Film with Primer

MCF-770G (Type P)

Product

- **Copper Clad Film MCF-770G(Type P)**

Applications

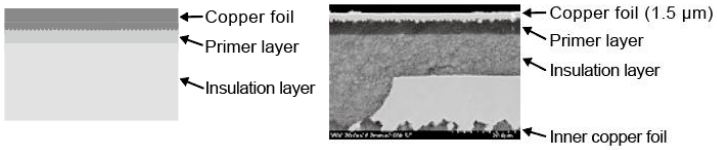
- **Semiconductor packages (FC-CSP, PoP, SiP)**
- **Thinner Module PWB**

Features

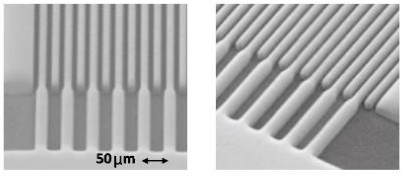
- Film material (copper clad film) that can achieve an insulation layer of 15 μm or less.
- Excellent warpage characteristics even in ultra-thin substrates by low CTE and high modulus
- Excellent adhesion with copper, enabling fine wiring formation by semi-additive process.
- Excellent insulation properties

Characteristics

Construction

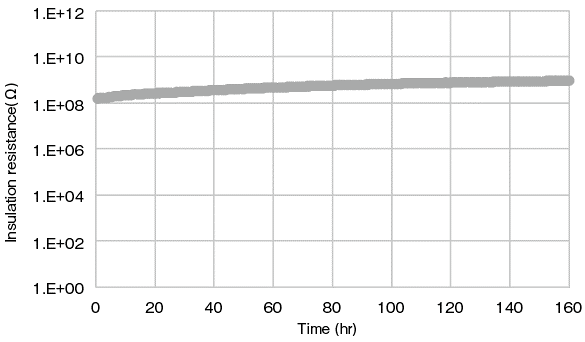


Fine patterning with Semi-additive process



Design
L/S = 10/10 μm with MCF-770G(Type P)

Reliability test (Layer to layer)



・Sample: MCF-200HS Dielectric thickness : 20 μm
・Pre-treatment condition: JEDEC
Level 2 (60 ℃/60 %RH 120hr + 260 ℃ reflow 6 cycle)
・Evaluation condition: 130 ℃/85 %RH, 6 V

Copper Clad Film

Item		Condition*3	Unit	Actual Value MCF-770G(Type P)	Reference (IPC-TM-650)
Tg	DMA	A	℃	290～310	–
CTE*1	X(30～120℃)	A	ppm/℃	18～20	–
	Y(30～120℃)			18～20	
Solder Heat Resistance(260℃)		A	sec.	≧ 300	–
Copper Peel Strength	3 μm	A	kN/m	0.9～1.0	2.4.8
	12 μm (LP)			0.9～1.0	
Elastic Modulus (Tensile)		A	GPa	8～10	–
Dielectric Constant	10 GHz*2	A	–	3.2～3.3	–
Dissipation Factor	10 GHz*2	A	–	0.006～0.008	–

*1) Heating Rate: 10℃/min. (Tensile) *2) Measured by Split post dielectric resonators (SPDR). *3) Refer to "Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

Copper Clad Film

Part Number	Copper foil type	Primer thickness	Dielectric thickness
MCF-770G(Type P)	1.5 μm, 3 μm, 12 μm (LP)	2 μm	15 ± 2 μm
			20 ± 2 μm
			25 ± 2 μm

High Tg, Low Transmission Loss Laminate Material

MCL-LW-990(Type RFD)

Product

- **CCL MCL-LW-990(Type RFD)**
- **Prepreg GWA-990(Type D)**

Applications

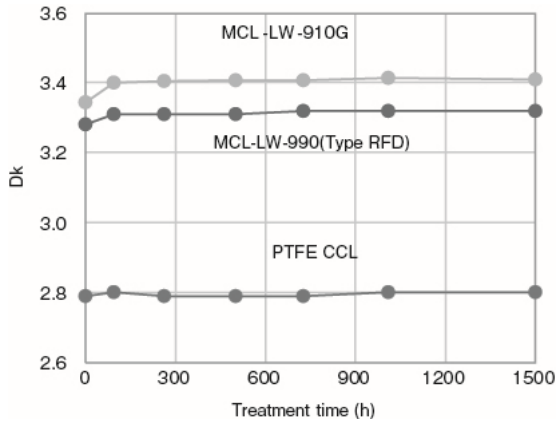
- **High frequency radar/antenna for automotive**
- **High frequency devices**

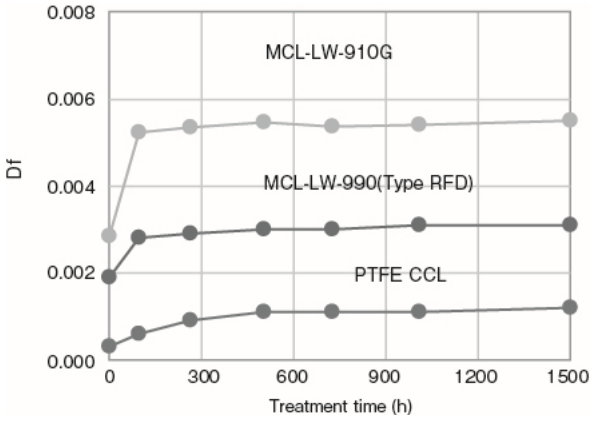
Features

- Achieved low dielectric constant (Dk) (3.3 at 10 GHz) and low dissipation factor (Df) (0.0020 at 10 GHz) using original resin system and low Dk glass cloth.
- Improved transmission loss by approximately 30% at 77 GHz compared to our MCL-LW-910G, combined with smooth copper foil.
- Has excellent heat resistance and connection reliability.

Characteristics

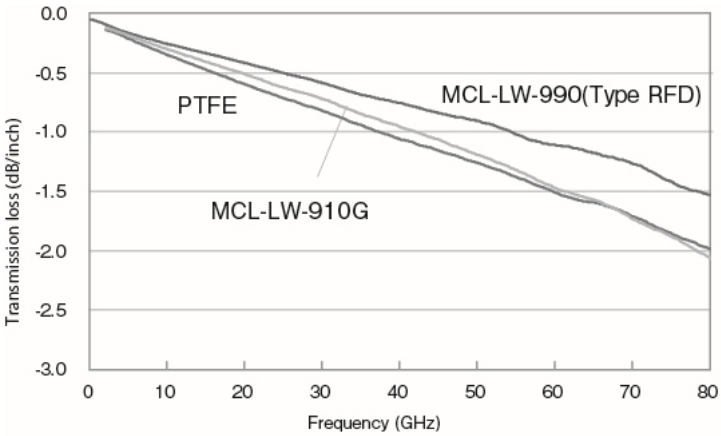
- **Dielectric Characterization Results (Aging test)**





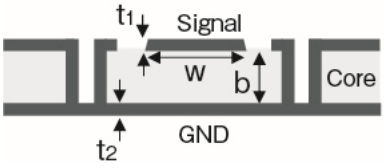
- Method : Cavity Resonator
- Aging condition (Temperature & Humidity) : 85℃/85%RH
- Measurement frequency : 10 GHz

- **Transmission Loss**



Measurement conditions

- Evaluation PWB : Micro Strip Line
- Temperature & Humidity : 25℃/60%RH
- Characteristic impedance : Approx. 50Ω
- Proofreading method : TRL(Thru-Reflect-Line)
- Trace width (w) : 0.3 mm
- Dielectric thickness (b) : 0.127 mm
- Trace thickness (t1) : 30 μm
- GND thickness (t2) : 30 μm
- Copper foil
  MCL-LW-990(Type RFD) : HVLP
  MCL-LW-910G : HVLP
  PTFE : Standard



●Copper Clad Laminate

(t0.8 mm)

Item			Condition*3	Unit	Actual Value	Reference (IPC-TM-650)
					MCL-LW-990(Type RFD)	
Tg	TMA method		A	℃	180～230	2.4.24
	DMA method				220～260	—
CTE*1	X(30～120℃)		A	ppm/℃	7～10	—
	Y(30～120℃)				7～10	
	Z	(＜Tg)	A		35～55	2.4.24
		(＞Tg)			240～290	
Solder Heat Resistance(260℃)			A	sec.	≧300	—
T-260(Without Copper)			A	min.	≧60	2.4.24.1
T-288(Without Copper)					≧60	
Decomposition Temperature (TGA method 5% Weight Loss)			A	℃	375～405	2.3.40
Copper Peel Strength	18 μm HVLP		A	kN/m	0.4～0.6	2.4.8
Flexural Modulus(Lengthwise)			A	GPa	14～19	2.4.4
Dielectric Constant	10 GHz*2		A	—	3.1～3.3	IEC-62810
Dissipation Factor	10 GHz*2		A	—	0.0018～0.0028	
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}	2.5.7
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}	
Insulation Resistance			A	Ω	1×10^{13}～1×10^{15}	—
			D-2/100		1×10^{12}～1×10^{14}	—

*1)Heating Rate: 10℃/min. (Comp.) *2)Measured by Cavity Resonator. *3)Refer to "Condition Note"
*Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-LW-990	(RFD)	12, 18, 35 μm (HVLP) 18, 35, 70 μm (RTF)	M0.06	0.06 mm
			M0.08	0.08 mm
			0.1	0.10 mm
			0.13	0.13 mm
			0.26	0.25 mm
			0.51	0.51 mm

Note1) Laminate thickness means dielectric layer thickness.

●Prepre

Part Number	Type		Glass Cloth	Properties	
			Style	Resin Content (%)	Dielectric Thickness after Lamination*1
GWA-990	0.05	(D1035N74)	1035	74±2	0.064 mm
	0.06	(D1078N66)	1078	66±2	0.078 mm
	0.06	(D1078N73)	1078	73±2	0.101 mm
	0.06	(D1078N75)	1078	75±2	0.112 mm
	0.08	(D3313N59)	3313	59±2	0.101 mm
Reference(IPC-TM-650)				2.3.16	-

*1) The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Halogen Free, High Tg, Low Transmission Loss Multilayer Material

MCL-LW-900G /910G

Product

- **CCL MCL-LW-900G/910G**
- **Prepreg GWA-900G/910G**

Applications

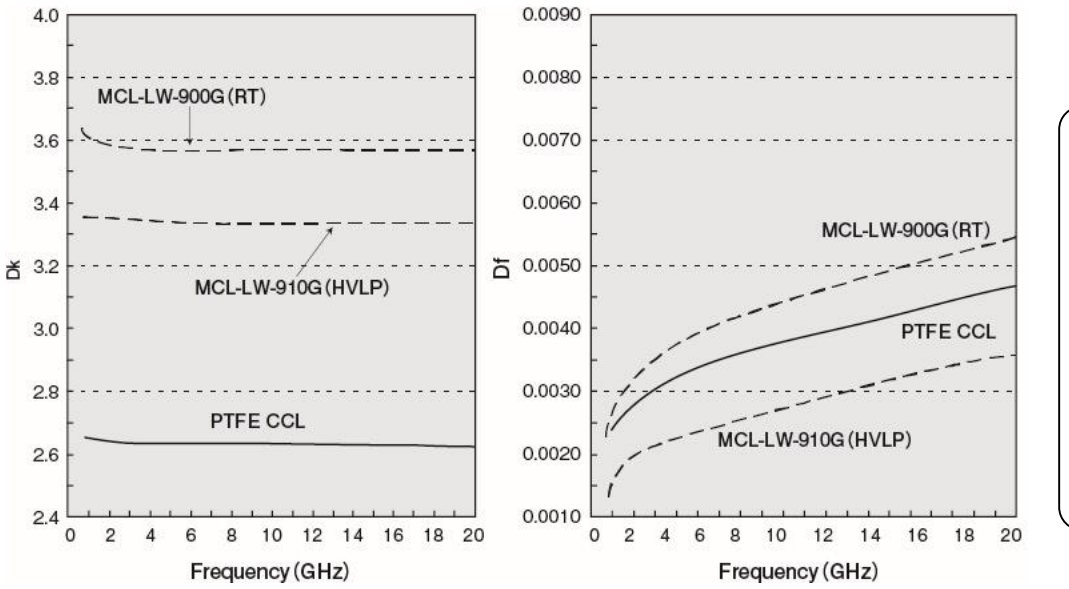
- **High-speed computer, server**
- **High-speed router**
- **High frequency devices, antenna**

Features

- MCL-LW-910G achieved low dielectric constant（3.3 at 10GHz）
  and low dissipation factor（0.0028 at 10 GHz）using low Dk glass and HVLP copper.
- Enables high-speed transmission/communication at 25 Gbps by low transmission loss properties.
- Has excellent heat resistance and connection reliability.

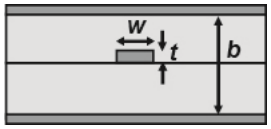
Characteristics

- **Dielectric characterization results**

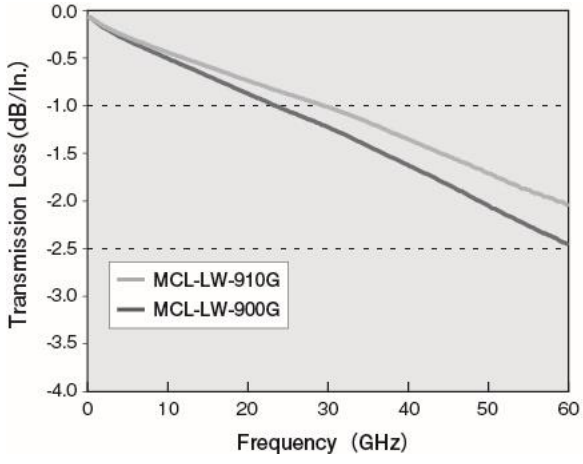


《Measurement Conditions》

- Method：Triplate-Line Resonator（JPCA-TM001）
- Temperature & Humidity：25℃/60%RH
- Laminate Thickness（b）：1.6 mm（Signal-Ground：800 μm）,Trace thickness（t）：18 μm（RT,HVLP）
- Signal Conductor Line Width（w）：1mm（Zo:ca.50Ω）

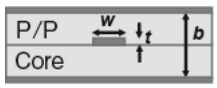


- **Transmission Loss**



《Measurement conditions》

- Evaluation PWB：Strip-line
- Temperature & Humidity：25℃/60%RH
- Characteristic impedance：Approx. 50Ω
- Inner layer surface treatment：Black-reduction
- Proofreading method：TRL（Thru-Reflect-Line）



- Trace width（w）：0.120 mm
- Dielectric thickness（b）：0.25 mm
- Trace thickness（t）：18 μm

●Copper Clad Laminate

（t0.8 mm）

Item			Condition*3	Unit	Actual Value		Reference（IPC-TM-650）
					MCL-LW-900G	MCL-LW-910G	
Tg	TMA method		A	℃	190～210		2.4.24
	DMA method		A		220～260		—
CTE*1	X（30 ～ 120℃）		A	ppm/℃	12～15		—
	Y（30 ～ 120℃）				12～15		
	Z	（＜Tg）	A		35～45		2.4.24
		（＞Tg）	A		230～280		
Solder Heat Resistance（260℃）			A	sec.	≧300		2.4.24.1
T-260（Without Copper）			A	min.	≧60		
T-288（Without Copper）					≧60		
Decomposition Temperature（TGA method 5% Weight Loss）			A	℃	410～450		2.3.40
Copper Peel Strength	18 μm RT		A	kN/m	0.4～0.7		2.4.8
	18 μm HVLP				0.4～0.7		
Flexural Modulus（Lengthwise）			A	GPa	16～21		2.4.4
Dielectric Constant	10 GHz*2		A	—	3.6～3.8	3.2～3.4	IEC-62810
Dissipation Factor	10 GHz*2		A	—	0.0040～0.0050	0.0025～0.0035	
Volume Resistivity			C-96/40/90	Ω･cm	1×10^{14}～1×10^{16}		2.5.17
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}		
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}		—
			D-2/100		1×10^{12}～1×10^{14}		—

*1）Heating Rate: 10℃/min. (Comp.)　*2）Measured by Cavity Resonator.　*3）Refer to “Condition Note”

※Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-LW-900G（E glass） MCL-LW-910G（Low Dk glass）	—	18μm 35μm 70μm （RT） 12μm 18μm 35μm （HVLP）	M0.05	0.05 mm
			M0.06	0.06 mm
			M0.08	0.08 mm
			0.1	0.10 mm
			M0.11	0.10 mm
			0.13	0.13 mm
			M0.15	0.15 mm
			0.2	0.20 mm
			0.26	0.25 mm

Note1) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type		Glass Cloth	Properties	
			Style	Resin Content（%）	Dielectric Thickness after Lamination*1
GWA-900G	0.05	（1037N72）	1037	72 ±2	0.050 mm
	0.06	（1078N65）	1078	65 ±2	0.078 mm
	0.08	（3313N57）	3313	57 ±2	0.106 mm
	0.1	（2116N55）	2116	55 ±2	0.125 mm
GWA-910G	0.05	（1037N74）	1037	74 ±2	0.050 mm
	0.06	（1078N67）	1078	67 ±2	0.078 mm
	0.08	（2013N59）	2013	59 ±2	0.106 mm
	0.1	（2116N57）	2116	57 ±2	0.125 mm

Reference（IPC-TM-650）	2.3.16	—

*1）The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Halogen Free, Low Dielectric Constant, High Heat Resistance Multilayer Material

MCL-HE-679G（Type S）

Product

- **CCL MCL-HE-679G (Type S)**
- **Prepreg GHA-679G (Type S)**

Applications

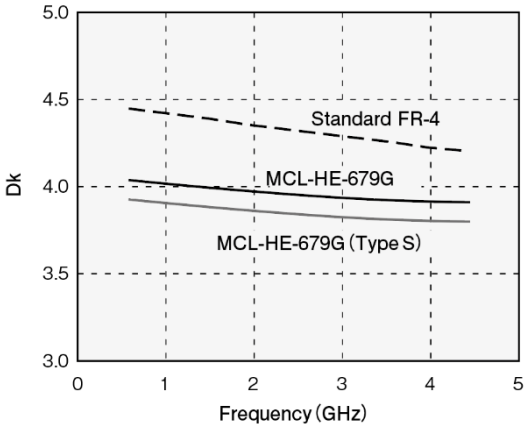
- **Network applications**
- **High-frequency parts（Filters, VCOs, etc.）**

Features

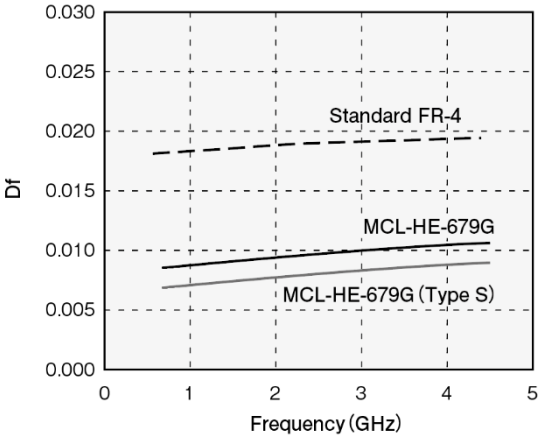
- Enables lower transmission loss than MCL-HE-679G with lower dissipation factor.
- High Tg and superior heat resistance for soldering.（Suitable for lead free process）
- CTE in Z-direction is 30% lower than that of our standard FR-4
- Achieved UL 94V-O flammability standard without using any compound which includes halogen, antimony or red phosphorous.

Characteristics

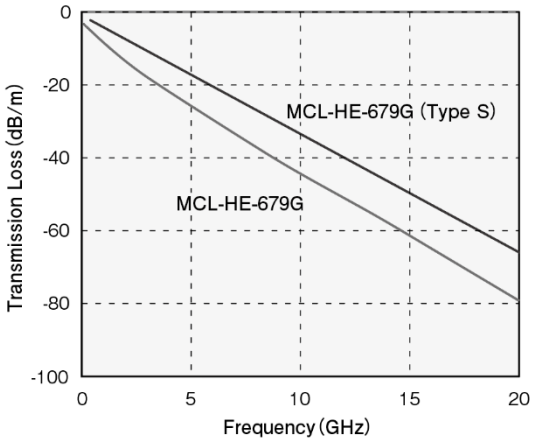
- **Correlation between Dielectric Constant and Frequency**



- **Correlation between Dissipation Factor and Frequency**



- **Transmission Loss**



●Copper Clad Laminate

（t0.8 mm）

Item			Condition*4	Unit	Actual Value	Reference（IPC-TM-650）
					MCL-HE-679G（Type S）	
Tg	TMA method		A	℃	180 ～ 190	2.4.24
	DMA method		A		260 ～ 280	—
CTE*1	X（30 ～ 120℃）		A	ppm/℃	12 ～ 15	—
	Y（30 ～ 120℃）				14 ～ 17	
	Z	（＜Tg）	A		30 ～ 40	2.4.24
		（＞Tg）	A		190 ～ 230	
Solder Heat Resistance（260℃）			A	sec.	≧300	—
T-260（Without Copper）			A	min.	≧60	2.4.24.1
T-288（Without Copper）					≧60	
Decomposition Temperature（TGA method 5% Weight Loss）			A	℃	370 ～ 390	2.3.40
Copper Peel Strength	18 μm		A	kN/m	0.5 ～ 0.7	2.4.8
	35 μm				0.6 ～ 0.8	
Flexural Modulus（Lengthwise）			A	GPa	23 ～ 26	2.4.4
Dielectric Constant	1 GHz*2		A	—	3.7 ～ 3.9	JPCA TM-001
	10 GHz*3				3.6 ～ 3.8	
Dissipation Factor	1 GHz*2		A	—	0.006 ～ 0.008	
	10 GHz*3				0.008 ～ 0.010	
Volume Resistivity			C-96/40/90	Ω·cm	1×10^{14}～1×10^{16}	2.5.17
Surface Resistance				Ω	1×10^{13}～1×10^{15}	
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}	—
			D-2/100		1×10^{12}～1×10^{14}	—

*1）Heating Rate: 10℃/min. (Comp.)　*2）Measured by Triplate-line Resonator.　*3）Measured by Material Analyzer.　*4）Refer to “Condition Note”

※Above data are experimental results and not guaranteed.

Standard Specifications

●Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-HE-679G	（S）	12 μm (STD) 18 μm 35 μm 70 μm (RT)	0.06	0.06 mm
			0.08	0.08 mm
			0.1	0.10 mm
			0.15	0.15 mm
			0.2	0.20 mm
			0.4	0.40 mm
			0.6	0.60 mm
			0.8	0.80 mm

Note1) Laminate thickness means dielectric layer thickness.

●Prepreg

Part Number	Type		Glass Cloth	Properties	
			Style	Resin Content（%）	Dielectric Thickness after Lamination*1
GHA-679G	0.04	（S1037N72）	1037	72±2	0.053 mm
	0.06	（S1080N64）	1080	64±2	0.080 mm
	0.06	（S1080N69）	1080	69±2	0.095 mm
	0.06	（S1078N64）	1078	64±2	0.078 mm
	0.08	（S3313N56）	3313	56±2	0.105 mm
	0.08	（S3313N62）	3313	62±2	0.126 mm
	0.1	（S2116N54）	2116	54±2	0.128 mm
	0.1	（S2116N60）	2116	60±2	0.152 mm
Reference（IPC-TM-650）				2.3.16	—

*1）Dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Polyimide Multilayer Material

MCL-I-671

Product

- **CCL MCL-I-671**
- **Prepreg GIA-671N**

Applications

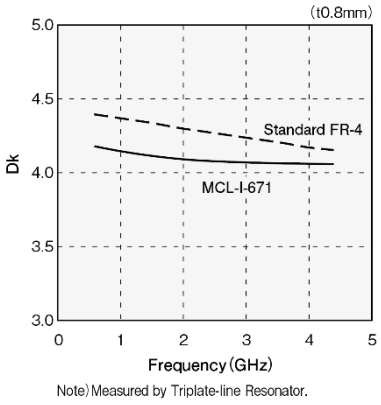
- **Main frame computers and super computers**
- **Semiconductor testing equipment, and burn-in boards**
- **Flex-rigid PWBs（No-flow prepreg）**

Features

- High Tg（＞200℃ : DMA）material for high through-hole reliability.
- FR-4 lamination condition can be applied.（175℃, 90 min.）
- MDA, a general polyimide resin curing agent, is not used.（Non-MDA Resin System）
- Resin flow control technology enables from high to low flow.

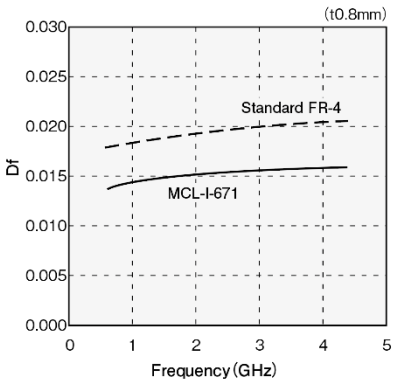
Characteristics

Correlation between Dielectric Constant and Frequency

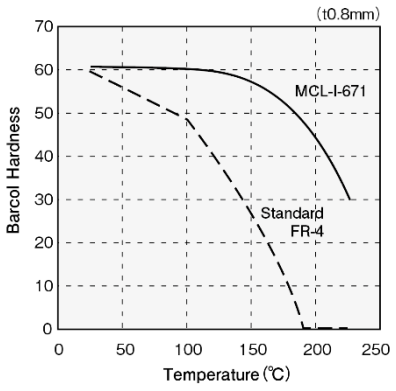


Note) Measured by Triplate-line Resonator.

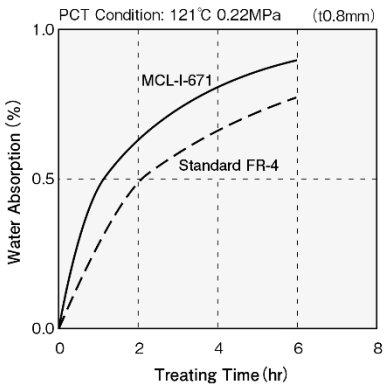
Correlation between Dissipation Factor and Frequency



Barcol Hardness

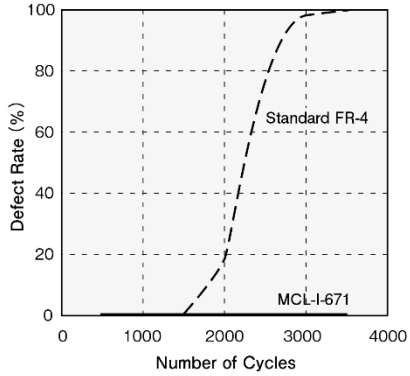


Water Absorption

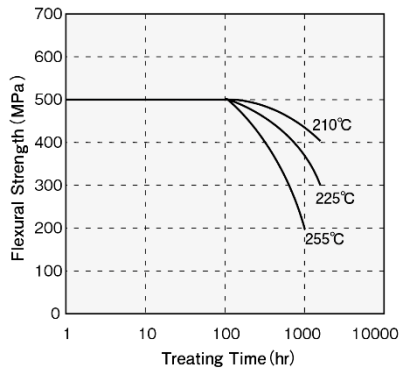


Thermal Shock Test MIL-STD-202

Method 107E



Flexural Strength



● Copper Clad Laminate

（t0.8mm）

Item			Condition*3	Unit	Actual Value MCL-I-671	Reference (IPC-TM-650)
Tg	TMA method		A	℃	200～213	2.4.24
	DMA method		A		230～245	—
CTE*1	X（30 ～ 120℃）		A	ppm/℃	12～15	—
	Y（30 ～ 120℃）				12～16	
	Z	（＜Tg）	A		50～80	2.4.24
		（＞Tg）	A		200～300	
Solder Heat Resistance（260℃）			A	sec.	≧300	—
T-260（Without Copper）			A	min.	≧60	2.4.24.1
T-288（Without Copper）					≧15	
Decomposition Temperature（TGA method 5% Weight Loss）			A	℃	330～350	2.3.40
Copper Peel Strength（18 μm）	20℃		A	kN/m	1.3～1.5	2.4.8
	180℃				1.0～1.2	
Flexural Modulus（Lengthwise）			A	GPa	24～26	2.4.4
Dielectric Constant	1 MHz		A	—	4.2～4.4	2.5.5.9
	1 GHz*2				4.1～4.3	JPCA TM-001
Dissipation Factor	1 MHz		A	—	0.0110～0.0130	2.5.5.9
	1 GHz*2				0.0130～0.0150	JPCA TM-001
Volume Resistivity			C-96/40/90	Ω･cm	1×10^{14}～1×10^{16}	2.5.17
Surface Resistance				Ω	1×10^{13}～1×10^{15}	
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}	—
			D-2/100		1×10^{12}～1×10^{14}	—

*1）Heating Rate: 10℃/min. (Comp.)　*2）Measured by Triplate-line Resonator.　*3）Refer to “Condition Note”

※Above data are experimental results and not guaranteed.

Standard Specifications

● Copper Clad Laminate

Part Number	Type	Copper Foil Thickness	Thickness Code	Laminate Thickness
MCL-I-671	—	12 μm 18 μm 35 μm 70 μm	0.06	0.06 mm
			0.1	0.10 mm
			0.2	0.20 mm
			0.3	0.30 mm
			0.4	0.40 mm
			0.8	0.80 mm
			1.0	1.00 mm
			1.2	1.20 mm
			1.6	1.60 mm

Note1) Laminate thickness means dielectric layer thickness.

● Prepreg

Part Number	Type		Glass Cloth Style	Properties Resin Content (%)	Properties Dielectric Thickness after Lamination*1	Application
GIA-671N	（T）	0.03	106	72±3	0.059 mm	MLB
		0.05	1080	65±3	0.087 mm	
		0.1	2116	54±3	0.134 mm	
	（N）	0.03	106	68±3	0.047 mm	Flex-rigid PWBs
		0.05	1080	59±3	0.072 mm	
	（F）	0.05	1080	74±3	0.111 mm	Metal core PWBs
Reference（IPC-TM-650）				2.3.16	—	—

*1）The dielectric thickness after lamination is defined as the theoretical calculation thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Reduction of Solder Crack/Low Elastic Modulus Material

TD-002

Product

- **Prepreg TD-002**

Applications

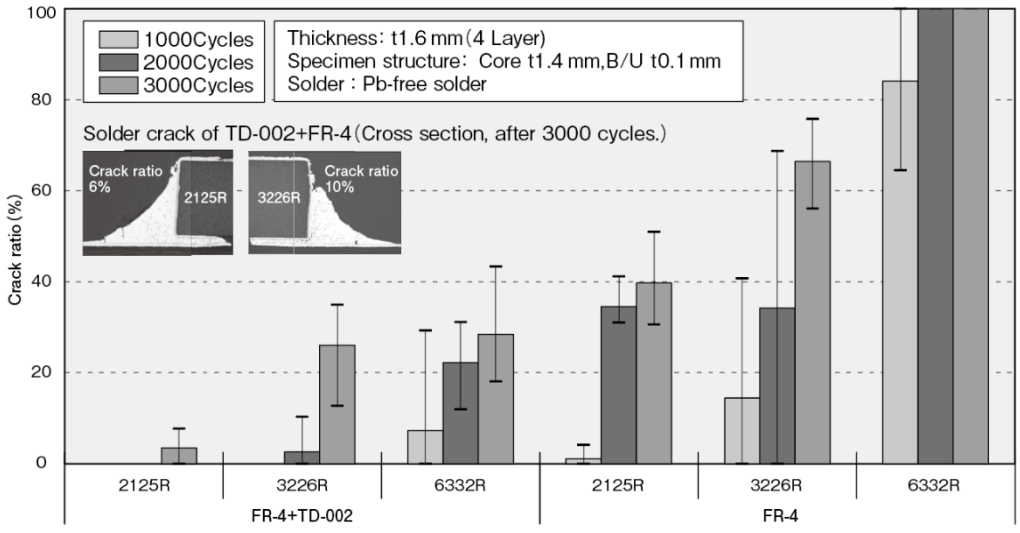
- **Electronic equipment for automobiles.**
- **Engine room set board.**
- **PCB on component packagings.**

Features

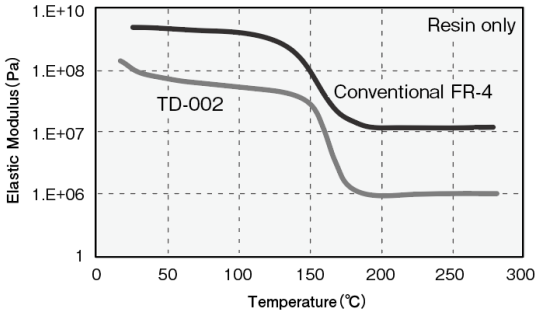
- Applied to PWB surface of standard material to absorb soldering stress and inhibit solder crack.
- Elastic modulus is 1/4 of that of standard FR-4.
- Enables to reduce solder crack without using high functional material by combining TD-002 with standard material.

Characteristics

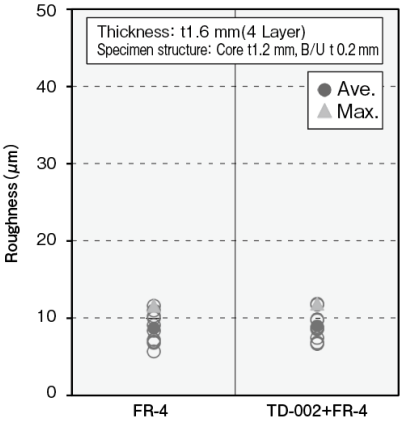
- **Connection reliability of Pb-free solder 〔Thermal condition：-40℃（30 min.）⇔125℃（30 min.）〕**



- **Elastic modulus**



- **Mechanical drilling processing〔After 3000 hits〕**



Appearance after drilling



TD-002+FR-4

《Condition of mechanical drilling》

- Stack-up：3 panels
- Revolution：120 krpm
- Drill bit：φ0.3 mm
- E/B：Al t0.15 mm
- Feeding Speed：2.4 m/min.

●Prepreg

（3313N58, t0.4 mm, t0.1 mm）

Item			Condition*4	Unit	Actual Value	Reference
					TD-002	（IPC-TM-650）
Tg	TMA method		A	℃	155～170	2.4.24
CTE*1	X*（*2 30 ～ 120℃）		A	ppm/℃	6～9	—
	Y*（*2 30 ～ 120℃）				6～9	
	Z	（＜Tg）	A		80～130	2.4.24
		（＞Tg）	A		200～300	
Solder Heat Resistance（260℃）			A	sec.	≧300	—
T-260（Without Copper）			A	min.	≧50	2.4.24.1
T-288（Without Copper）					≧5	
Decomposition Temperature （TGA method 5% Weight Loss）			A	℃	345～360	2.3.40
Copper Peel Strength	18μm		A	kN/m	0.8～0.9	2.4.8
	35μm				0.9～1.1	
Flexural Modulus（Lengthwise）			A	GPa	5～8	2.4.4
Elastic Modulus（Lengthwise）			A	GPa	7～10	—
Dielectric Constant	1 GHz*3		A	—	3.6～3.8	IEC-62810
Dissipation Factor	1 GHz*3		A	—	0.011～0.013	
Volume Resistivity			C-96/40/90	Ω•cm	1×10^{14}～1×10^{16}	2.5.17
Surface Resistance			C-96/40/90	Ω	1×10^{13}～1×10^{15}	
Insulation Resistance			A	Ω	1×10^{14}～1×10^{16}	—
			D-2/100		1×10^{12}～1×10^{14}	

*1）Heating Rate: 10℃/min.　*2) Tensile　*3）Measured by cavity resonator.　*4）Refer to "Condition Note"

※Above data are experimental results and not guaranteed.

Standard Specifications

●Prepreg

Part Number	Type		Glass Cloth	Properties	
			Style	Resin Content （%）	Dielectric Thickness after Lamination*1
TD-002	0.06	（1037N77）	1037	77±2	0.069 mm
	0.08	（1078N66）	1078	66±2	0.088 mm
	0.10	（3313N58）	3313	58±2	0.115 mm

Reference（IPC-TM-650）	2.3.16	—

*1）The dielectric thickness after lamination is defined as the thickness of one sheet of prepreg when the resin flow is 0%.
Values change depending on the press condition or inner layer pattern.

Copper Foil for Fine Patterning

PF-EL

Product

- **Copper Foil PF-EL**

Applications

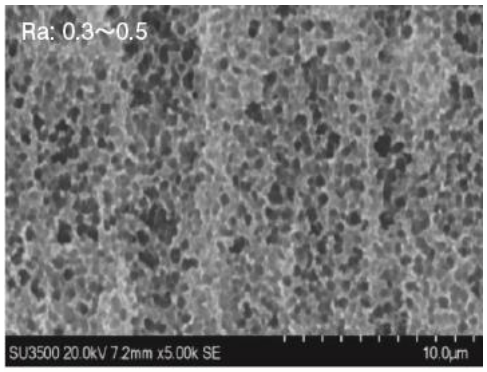
- **Semiconductor package substrates**
- **High density multi-layer PWB**

Features

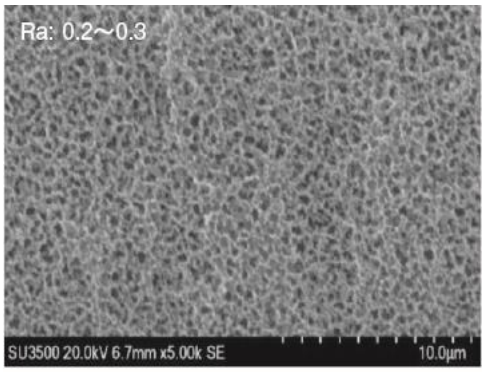
- PF-EL is a copper foil that is appropriate for fine line patterning with semi-additive process（SAP）using rough shapes of primer made with the copper profile.
- PF-EL has high peel strength for plating copper.
- High flexural modulus substrates with using prepregs.

Characteristics

- **Surface**

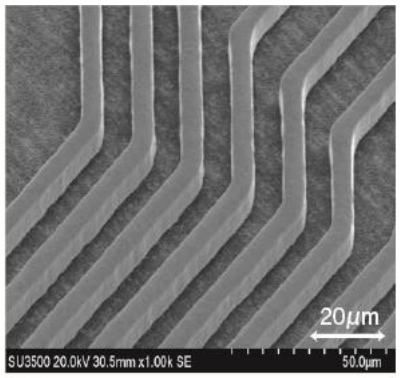


PF-EL

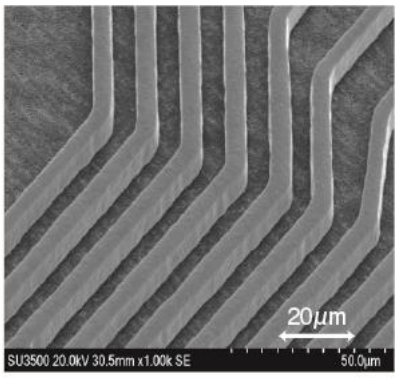


PF-EL SP

- **Fine patterning with SAP**



Design
L/S=10/10μm with PF-EL
(Exposure LDI)

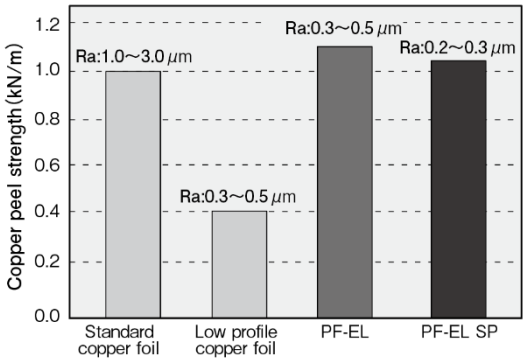


Design
L/S=7/7μm with PF-EL SP
(Exposure LDI)



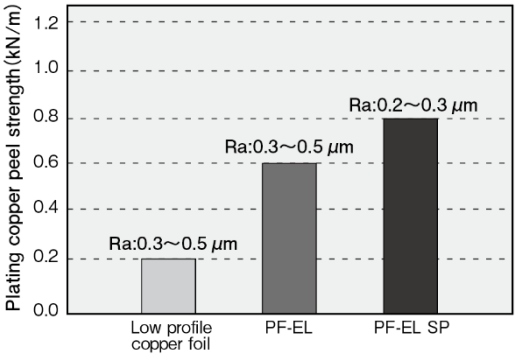
Design
L/S=5/5μm with PF-EL SP
(Exposure Stepper)

●Copper peel strength



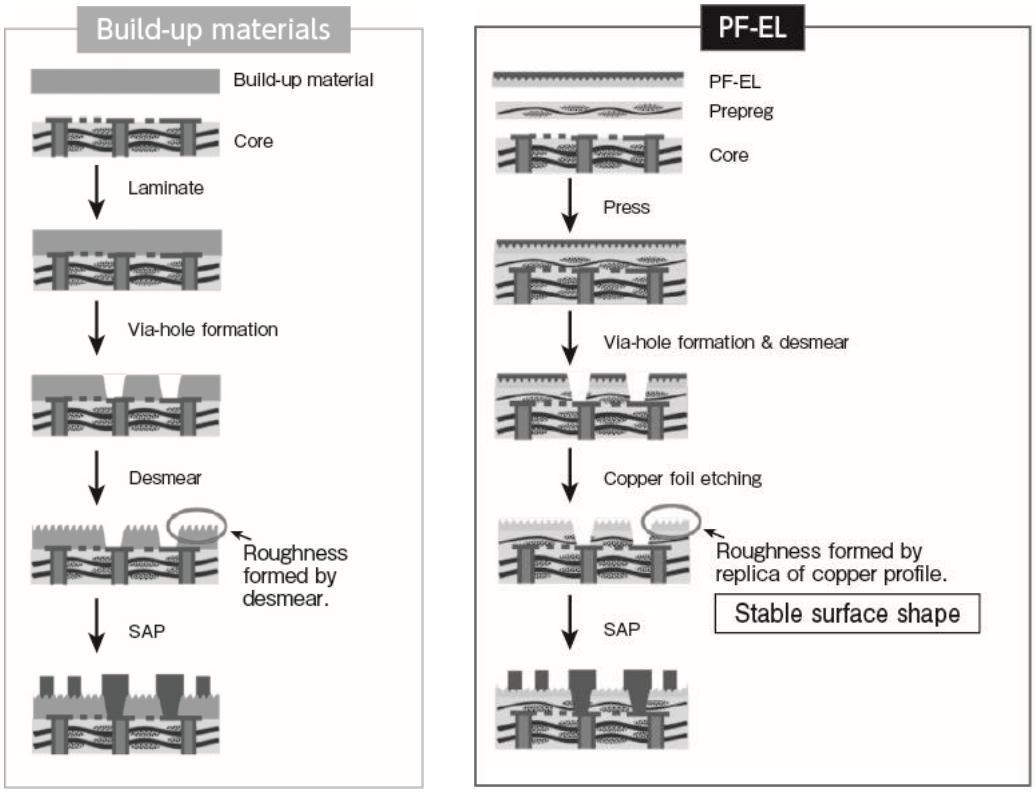
※MCL-E-770G(Type R), Copper foil 1.5 μm with plating copper 20 μm

●Plating copper peel strength



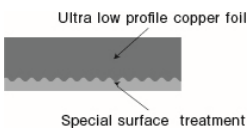
※MCL-E-770G(Type R), plating copper thickness 20 μm

●Fine patterning process



Standard Specifications

●Primer with Copper Foil

Part Number	Copper Foil Thickness (μm)	Special Surface Treatment (μm)	Roughness (μm)	Composition	Process
PF-EL-12	12	4	Ra:0.3 ～ 0.4 Rz:1.5 ～ 2.5		SAP*1
PF-EL-3	3	2,4	Ra:0.3 ～ 0.5 Rz:1.5 ～ 2.5		SAP*1 MSAP*2
PF-EL-2	2	2,4	Ra:0.3 ～ 0.5 Rz:1.5 ～ 2.5		
PF-EL-1.5SP	1.5	2	Ra:0.2 ～ 0.3 Rz:1.0 ～ 2.0		

*1) After lamination of the material to the prepreg,the copper foil is etched out,and the special surface treatment with appropriate roughness made by a replica of the copper profile etched out remains on prepreg surface.This process is SAP using this replica.

*2) Semi additive process uses thin copper foil as seed layer, and circuit pattern is formed by adding additional copper and then etching away that seed layer.

Precautionary Statement

Precautions for Use

● Handling

Take the following measures to avoid contacting and inhaling resin, glass or other dust from material, etc.

①Install a local ventilator above the workbench so as to sufficiently exhaust dust.
②Wear protection gear to cover hands, mouth, nose and eyes.
③Control room temperature not to sweat and protect skin from dust.

If you come into contact with and/or inhale resin, glass or other dust from material, take the following measures:

①If dust has entered into eyes : Wash your eyes with clean running water.
②If dust has come in contact with skin : Wash it off with soap water.
③If you have inhaled dust : Rinse out your mouth, or use tissues to clean if inhaled from nostrils.
④If you have ingested dust : Induce vomiting, or if it remains in your mouth, spit it out and then gargle.

Make sure to install a local ventilator since gas is generated in the heating process.

● Storage

Since material characteristics may change depending on the storage conditions, pay attention to the following points:

①Protect the materials from direct sunlight; otherwise, discoloration may occur.
②Use ventilator or air conditioner to remove moist air in the storage area and circulate the air.
③Keep the materials horizontally to prevent warpage or deflection and avoid piling materials on pallets.
Also, since condensation may occur on the surface of the walls, keep the materials away from walls to prevent them from becoming wet.
④Place dry paper or polyethylene film between materials to prevent them from getting scratched.
⑤Maintain constant room temperature to prevent from condensation into water on the materials.
⑥As prepregs and films require low-temperature storage, take sufficient time to get them back to ordinary temperature in packed condition. Before use, confirm there is no condensation on the surface.
（It takes about 12 hours to bring the temperature of materials stored at 5℃ to 22℃）.

● Storage Conditions

<table>
<tr><th colspan="2">Classification</th><th colspan="2">Part Number</th><th>Temperature</th><th>Humidity</th><th>Term</th><th>Precautions</th></tr>
<tr><td rowspan="15">Multilayer Material</td><td rowspan="7">CCL（Core Material）</td><td>MCL-E-795G</td><td>MCL-HS100</td><td rowspan="7">max. 25℃</td><td rowspan="7">50±15%RH</td><td rowspan="7">Within 6 months after shipment</td><td rowspan="16">Store in packed state</td></tr>
<tr><td>MCL-E-705G</td><td>MCL-HD60/90</td></tr>
<tr><td>MCL-E-700G</td><td>MCL-LW-990</td></tr>
<tr><td>MCL-E-770G</td><td>MCL-LW-910G</td></tr>
<tr><td>MCL-E-679FG</td><td>MCL-LW-900G</td></tr>
<tr><td>MCL-E-679FGB</td><td>MCL-HE-679G</td></tr>
<tr><td>MCL-HS200</td><td>MCL-I-671</td></tr>
<tr><td rowspan="8">Prepreg</td><td>GEA-795G</td><td>GD-60/90</td><td rowspan="8">20±5℃</td><td rowspan="8">50±10%RH</td><td rowspan="8">Within 3 months after shipment</td></tr>
<tr><td>GEA-705G</td><td>GWA-990</td></tr>
<tr><td>GEA-700G</td><td>GWA-910G</td></tr>
<tr><td>GEA-770G</td><td>GWA-900G</td></tr>
<tr><td>GEA-679FG</td><td>GHA-679G</td></tr>
<tr><td>GH-200</td><td>GIA-671N</td></tr>
<tr><td>GH-100</td><td></td></tr>
<tr><td colspan="2">TD-002</td><td>5±4%</td><td>50±10%RH</td><td>Within 2 months after shipment</td></tr>
<tr><td colspan="2" rowspan="2">Adhesive Film</td><td colspan="2">MCF-200HS</td><td>20±5℃</td><td>50±10%RH</td><td>Within 3 months after shipment</td></tr>
<tr><td colspan="2">MCF-770G(Type P)</td><td>20±5℃</td><td>50±10%RH</td><td>Within 2 months after shipment</td></tr>
</table>

Precautions in Process

- **Etching**

  Cupric chloride, ferric chloride or ammonium persulfate is generally used. After etching, wash the laminates carefully with clean running water. If washing is insufficient, residual ions (copper ions, iron ions, etc.) may degrade electric characteristics, lower copper foil adhesiveness and discolor laminates.

- **Removing etching resist**

  To prevent boards from discoloration or other change, minimize processing time.After processing,wash the materials well with water and then dry them thoroughly.

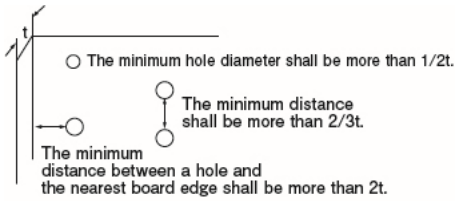
- **Punching**

  The following three principles are necessary for CCL punching.

  ①The minimum hole diameter shall be more than 1/2 of the board thickness.

  ②The minimum distance between holes shall be more than 2/3 of the board thickness.

  ③The minimum distance between a hole and the nearest board edge shall be more than twice the board thickness.



Three Principles of Punching

- **Drilling**

  Pay attention to the total stacked thickness (t) of the CCL to be drilled and the effective length of the drill is 1.5 mm or more longer than (t). A spade type is suitable for glass fabric-based.Since hole quality varies with the type of drill used, the drilling ma- chine and drilling conditions, examine these conditions carefully when setting or changing them.

- **Plating**

  Carefully control the composition and temperature of the plating solution and plating time when plating CCL. After taking them out of the plating bath, wash them carefully with running water immediately then dry them sufficiently.

- **Use of UV Ink**

  The surface state of CCL influences adhesion and wettability of the UV ink, so pay attention to the following points:

  ①Polish the CCL surfaces sufficiently.

  ②After polishing, wash them sufficiently with running water.

  ③Do not store them for a long time.

  ④If stored for a long time, polish again, wash with water and dry them before use.

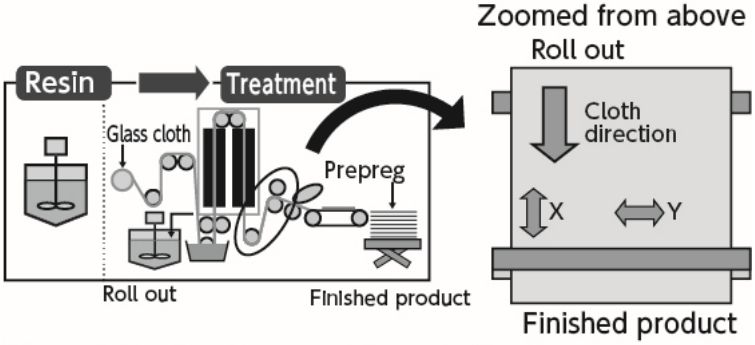
- **Heating**

  Heat CCL uniformly, not partially, as much as possible. When heating them to temperature higher than the glass-transition temperature, warpage and deflection can be reduced by using a horizontal rack.When heating them at high temperature for more than 30 minutes, the laminates may become discolored. So, determine the optimum time in advance.

- **Directionality of "CCL"**

  Since CCL is are directional（warp and fill）, be careful with respect to pattern design and lay out.

Production Process : Direction of CCL



Glass cloth direction = X (warp) direction

Precautions for Use of Completed PWB

● **Storage**

Since printed wiring boards absorb moisture in the environment, pay attention to the following points when storing them:

①Keep in a moisture-proof package (polyethylene bag and silica gel) until using them.

②Store in an air-conditioned room (below 25℃ and below 60%RH).

③Do not leave boards in a non-air-conditioned room (humidity: over 60%RH) before mounting.

④When using only some of the boards, keep the remaining ones in the same moisture-proof package in

⑤Even when kept in a moisture-proof package, these boards will gradually absorb moisture in the environment during humid seasons. When storing them for more than three months, take care to keep them dry.

● **Drying**

When boards are assumed to have absorbed moisture because of improper storage conditions or long-term storage of more than three months, be sure to dry them before use.

①Recommended drying conditions:

Drying temperature (C°)	Drying time (hrs.)
80	24
100	12
130	2

＊In case boards are stacked at drying, extend the drying time by about 20% because temperature of the boards rises slowly.

②Hold the boards horizontally during drying.

③Too high drying temperature (over 130℃) may cause oxidization of copper foil and discoloration of boards.

④In case of devices are mounted on boards, dry the boards at temperature at which the heat resistance of the part is taken into account. This applies to boards with surface protection coating.

⑤After drying, component mounting as soon as possible. If the work does not proceed smoothly, protect these boards with a moisture- proof package.

● **Heating**

①Note that excessive force on the board during. Heating may cause bow,twist or measling.

②During high temperature heat treatment (such as fusing), rapid heating and cooling may result in interlayer cracks, board swellings, and/or other defects. Such heating or cooling should be done step-by-step.To improve reliability, be sure to dry the materials before heat treatment.

● **Soldering**

The solder heat resistance of CCL is greatly affected by the temperature of the solder bath. Use an L-shaped thermometer to control carefully the temperature of the solder bath. When using a soldering iron, note that the actual temperature at the tip varies on its manufacturer and structure even though soldering irons are of the same capacity. You are recommended to measure the temperature with a surface thermometer in advance and to use the soldering iron below 300°C. After soldering, try not to move the substrates while the soldered portions remain hot.

Measling and blistering of the glass fabric epoxy CCL are apt to occur on wet printed wiring boards. So dry them (at 130℃ for 1 to 2 hours) before soldering.Excessively hot solder may result in shorter heat-resistance time, leading to swellings in circuits and boards. Make sure to maintain the appropriate temperature.

Recommended Press Condition for Multilayer Lamination

GEA-795G

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (80～130℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～3.5 MPa.)

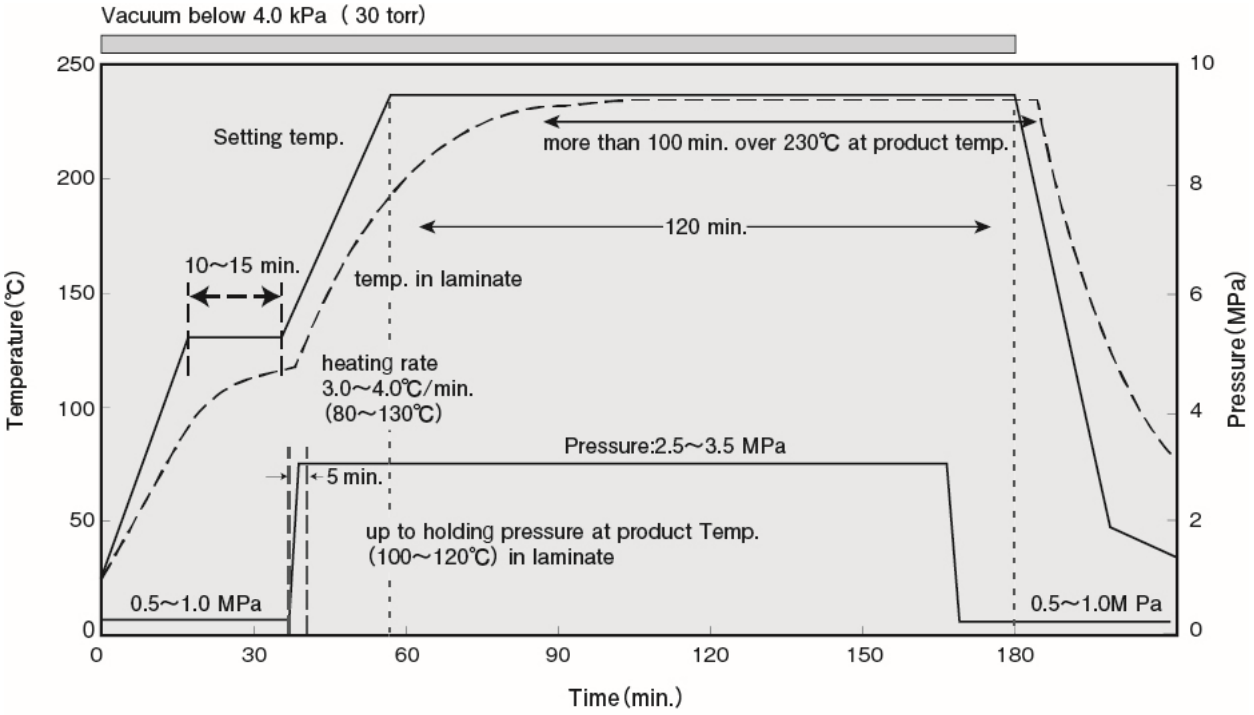
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 100～120℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-795G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-795G or GEA-795G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GEA-705G

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (80～130℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～3.5 MPa.)

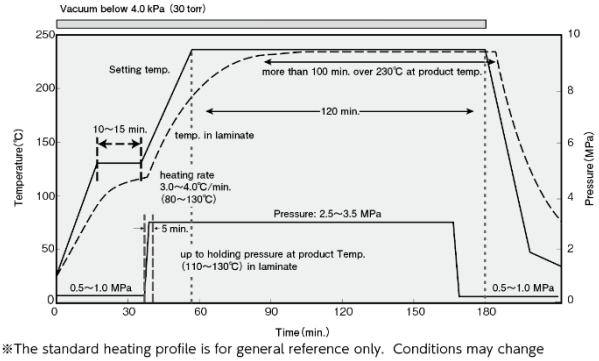
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-705G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-705G or GEA-705G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GEA-705G

TYPE-F

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 4.0～5.0℃/min. (<25 μm) or 3.0～4.0℃/min. (≧2.5 μm) within the melting temperature range of the prepreg resin (80 ～ 150℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 4.0~5.0 MPa for <25 μm, 2.0～3.0 Mpa for ≧25 μm)

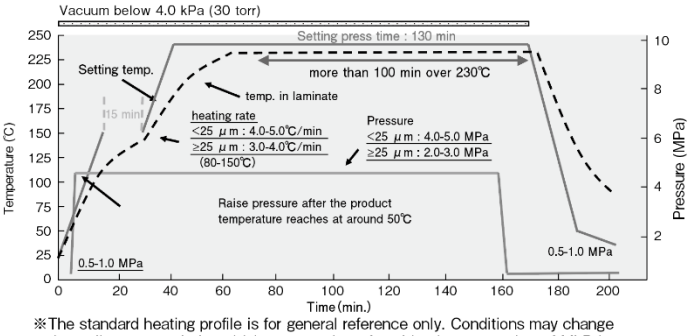
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-705G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-705G or GEA-705G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GEA-700G

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.0～3.0℃ /min. within the melting temperature range of the prepreg resin (80～150℃). Hold the product temperature above 220℃ for more than 80 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0～3.0 MPa.)

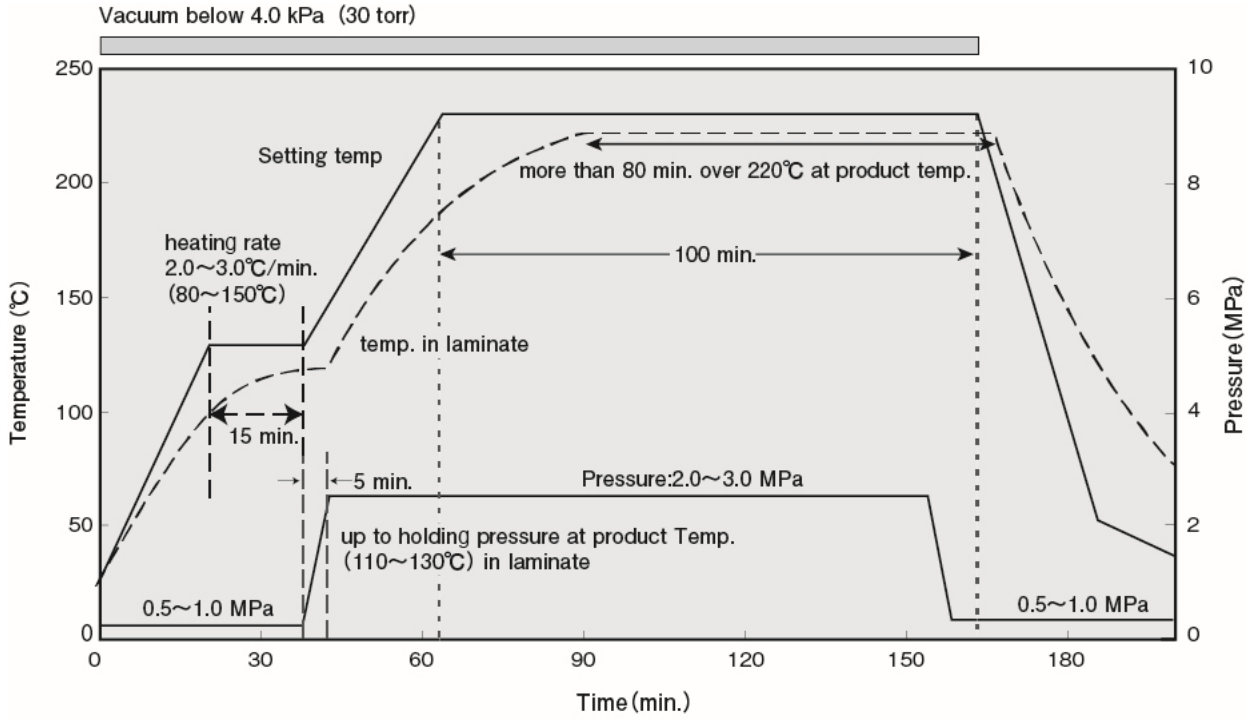
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-700G (Type R) as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-700G (Type R) or GEA-700G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GEA-770G

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (80～150℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0～3.0 MPa.)

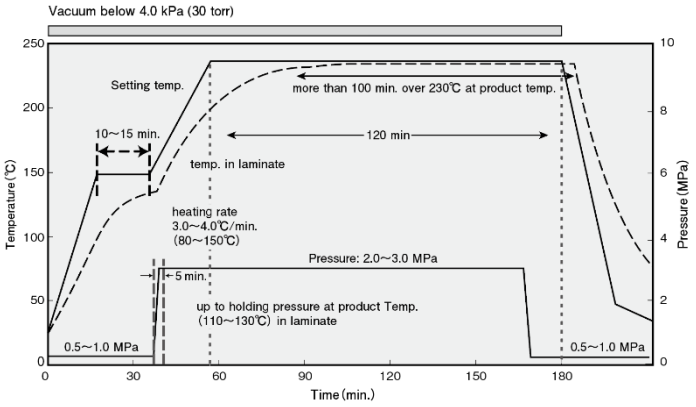
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-770G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-770G or GEA-770G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GEA-770G

TYPE-F

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 4.0～5.0℃/min. (<25 μm) or 3.0～4.0℃/min. (≧2.5 μm) within themelting temperature range of the prepreg resin (80～150℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 4.0~5.0 MPa for <25 μm, 2.0～3.0 Mpa for ≧25 μm)

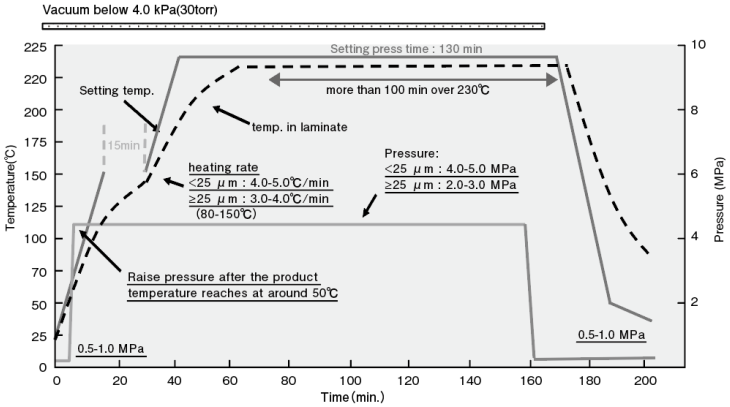
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-770G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-770G or GEA-770G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness,number of cushion layers, number of MLB lay-up, etc.

GEA-679FG

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be 1.0～2.5℃/min. within the melting temperature range of the prepreg resin (80～130℃). Hold the product temperature above 170℃ for more than 40 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa (30 torr). Please start the vacuuming before adding heat and pressure for lamination.

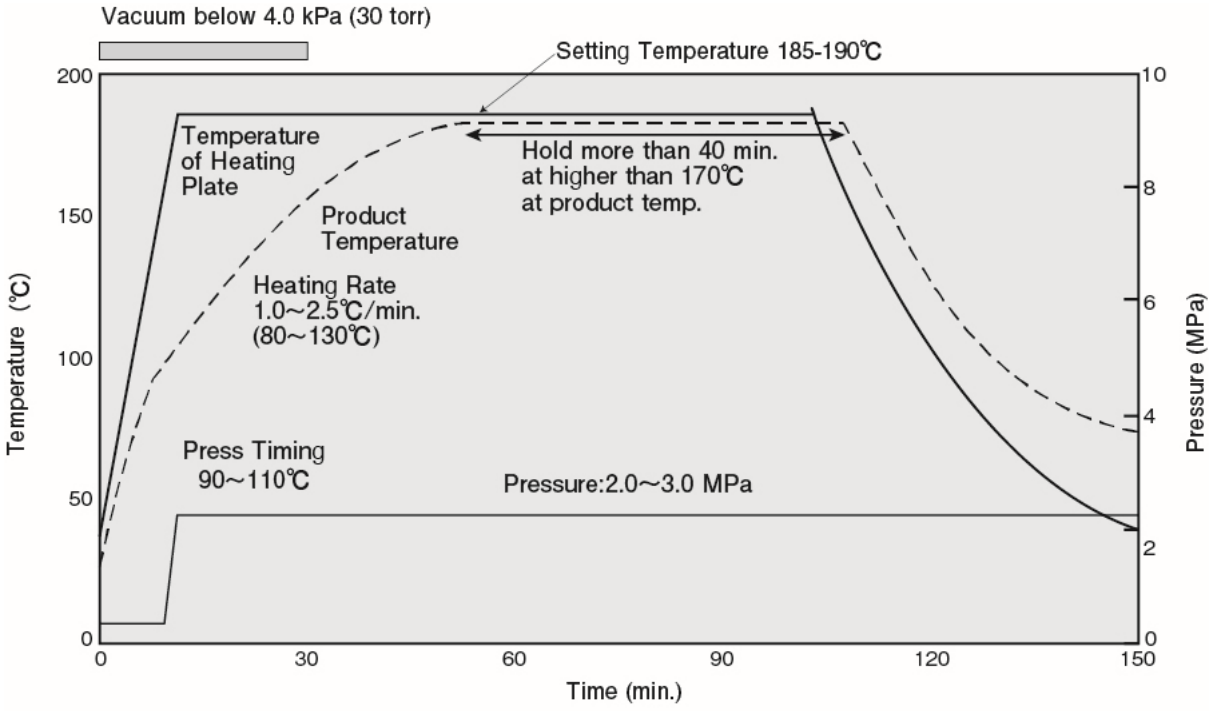
④The time for vacuuming shall not be longer than the end of molding process (When resin flow stops). If the vacuuming is continued until the end of the lamination process, flowed resin becomes fragile due to bubble remain. Fragments of it may become a cause of dent. Also small bubbles (blur) may occur in the edge of the product.

⑤The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0～3.0 MPa.)

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦In order to reduce the pressure shock to the product, it is recommended to set initial pressure low, 0.5 MPa with 90～110℃ product temperature at the initial stage.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GH-200

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.5～3.5℃/min. within the melting temperature range of the prepreg resin (50～160℃). Keep the product temperature at 225～235℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0kPa(30torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0～3.0 MPa.)

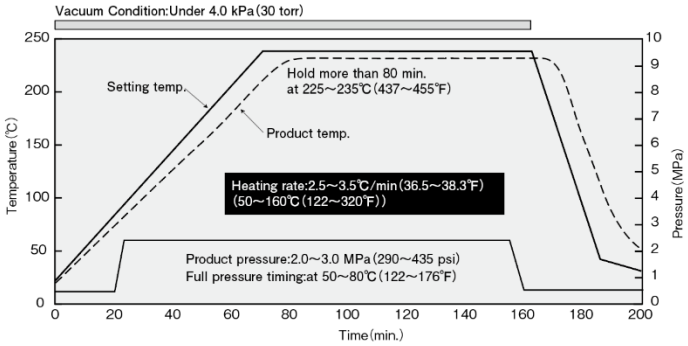
⑤We recommend that pressure on material is low pressure of 0.5MPa for resin of prepreg at the initial stage and set the full-pressure, before product temperature become 80℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By outgas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HS200 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HS200 or GH-200 for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GH-200

TYPE-F

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.0～3.0℃/min. within the melting temperature range of the prepreg resin (80～150℃). Keep the product temperature at 225～235℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0kPa(30torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0 MPa.)

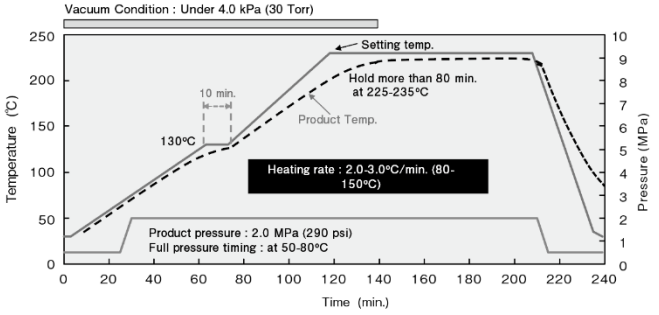
⑤We recommend that pressure on material is low pressure of 0.5MPa for resin of prepreg at the initial stage and set the full-pressure, before product temperature become 80℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By outgas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HS200 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HS200 or GH-200 for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GH-100

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.5～3.5℃/min. within the melting temperature range of the prepreg resin (80～150℃). Keep the product temperature at 195～205℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～3.5 MPa.)

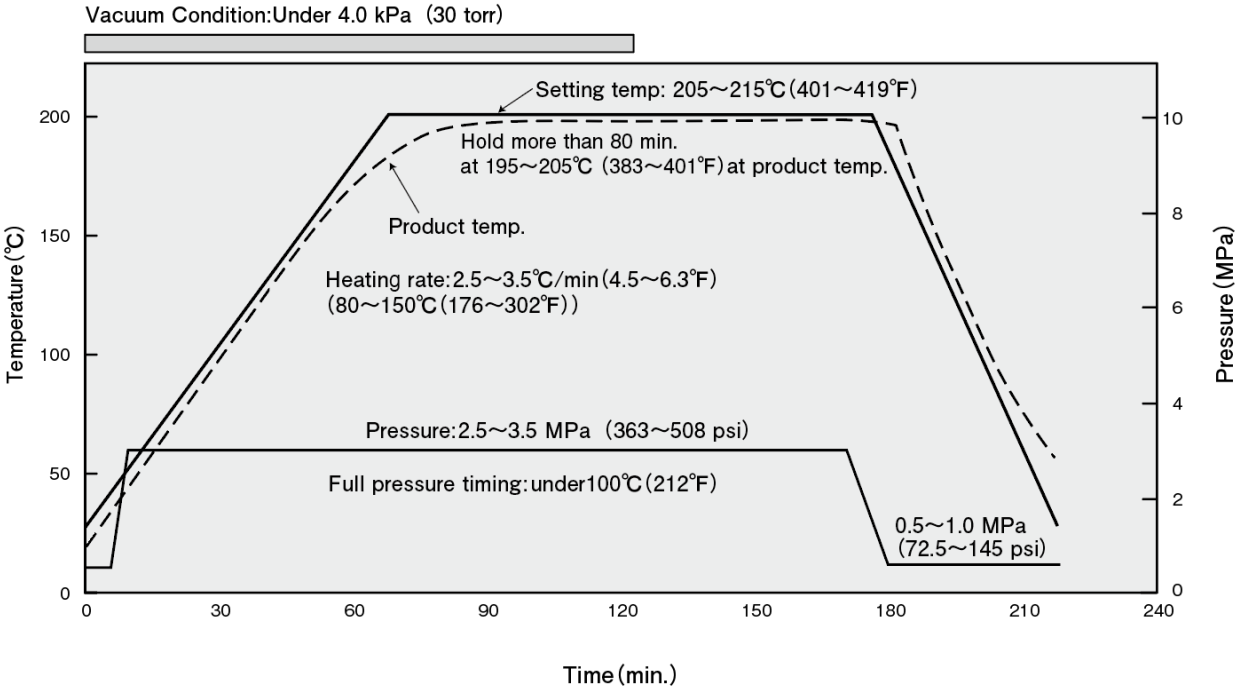
⑤We recommend that pressure on material is low pressure of 0.5 MPa for resin of prepreg at the initial stage and set the full-pressure,before product temperature become 100℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By outgas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HS100 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HS100 or GH-100 for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GD-60

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (50～160℃). Keep the product temperature at 225～235℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensionalchange, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 3.0～4.0 MPa.)

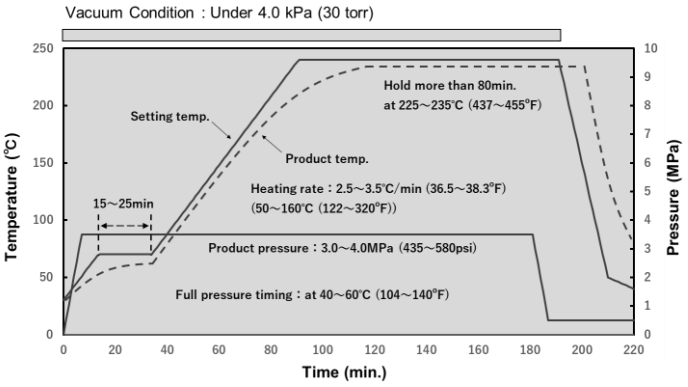
⑤We recommend that pressure on material is set the full-pressure before product temperature become 60 ℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HD60 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HD60 or GD-60 for dummy laminates, too.

● Recommended Press Conditions



GD-80

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (50～160℃). Keep the product temperature at 225～235℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensionalchange, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 3.0～4.0 MPa.)

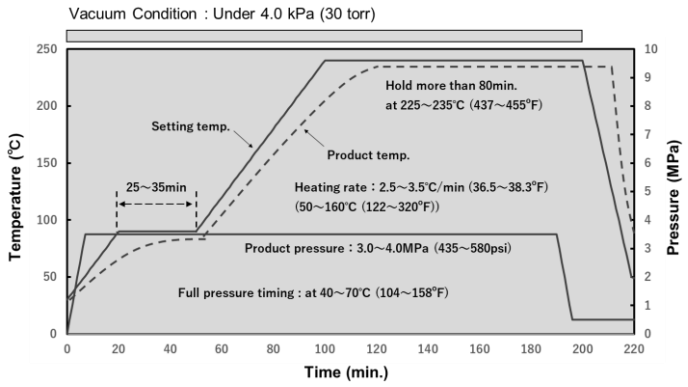
⑤We recommend that pressure on material is set the full-pressure, before product temperature become 70℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By outgas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HD90 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HD80 or GD-80 for dummy laminates, too.

● Recommended Press Conditions



GD-90

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (50～160℃). Keep the product temperature at 225～235℃ for more than 80 min.

②The setting temperature of heating plates differs depending on the heating rate or cushions, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensionalchange, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 3.0～4.0 MPa.)

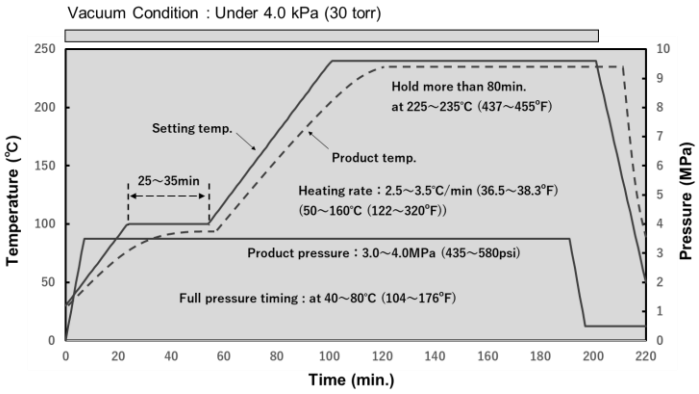
⑤We recommend that pressure on material is set the full-pressure, before product temperature become 80℃.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By outgas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HD90 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates decompose because of high lamination temp. Please use MCL-HD90 or GD-90 for dummy laminates, too.

● Recommended Press Conditions



MCF-200HS

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.5～3.5℃/min. within the melting temperature range of the prepreg resin (80～150℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～3.5 MPa.)

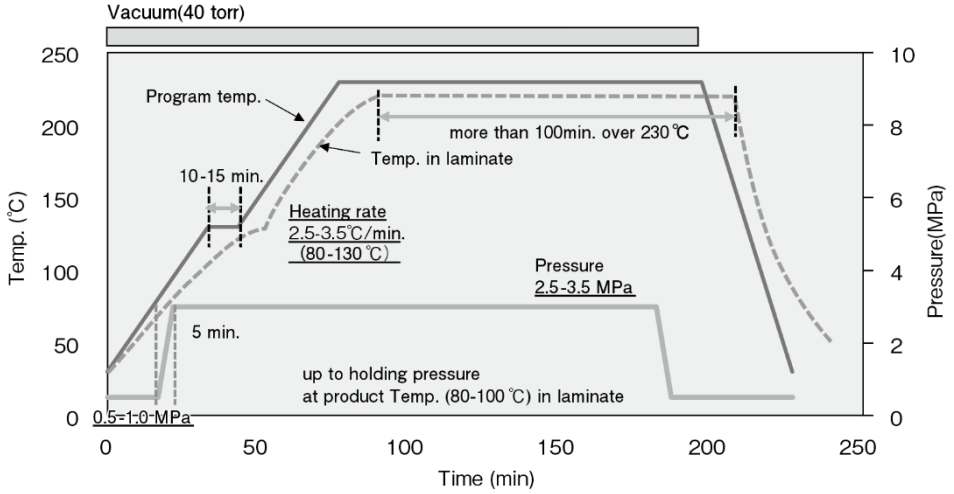
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 80～100℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-HS200 as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-HS200 or GH-200 for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

MCF-770G(Type P)

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 2.0～3.0℃/min. within the melting temperature range of the prepreg resin (80～150℃). Hold the product temperature above 230℃ for more than 100 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa(30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.0～3.0 MPa.)

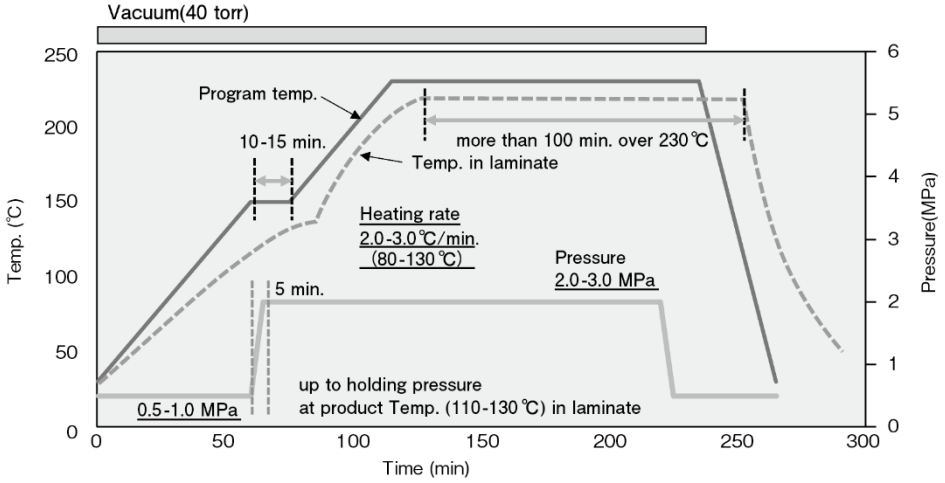
⑤In order to reduce the pressure shock to each part of the laminate, low pressure shall be applied to the material at the initial stage and increase it at the material temperature 110～130℃, when the resin becomes soft.

⑥Inner pattern filling capability of prepreg may change according to inner pattern and thickness design.

⑦By out gas from an inner substrate, the curing property of prepregs may be affected. Please use MCL-E-770G as an inner substrate.

⑧When you will use FR-4 laminates as dummy products, it is possible that FR-4 laminates will decompose because of high lamination temp. Please use MCL-E-770G or GEA-770G for dummy laminates, too.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GWA-990(Type D)

● Precautions for Outer Layer Copper Foil

Some copper foils have low peel strengths and are not suitable.
Please check copper peel strengths of the outer layer copper foil before use.

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (80℃～).

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

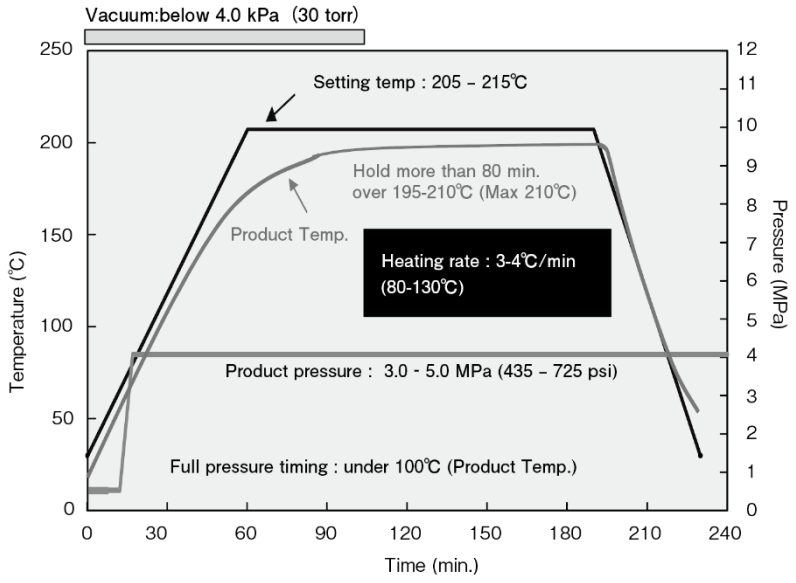
③The degree of vacuum shall be below 4.0 kPa (30 torr). Please start the vacuuming before adding heat and pressure for lamination.

④The time for vacuuming shall not be longer than the end of molding process. If the vacuuming is continued until the end of the lamination process, flowed resin becomes fragile due to bubble remain. Fragments of it may become a cause of dent. Also small bubbles (blur) may occur in the edge of the product.

⑤The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 3.0～5.0 MPa at vacuum pressing.)

⑥We recommend that pressure on material is the low pressure of 0.5 MPa for resin of prepreg at the initial stage and material temperature pressurizes it befor 80℃. The starting pressure is 0.5 MPa, and set the full-pressure, before product temperature become 100℃.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GWA-900G/GWA-910G

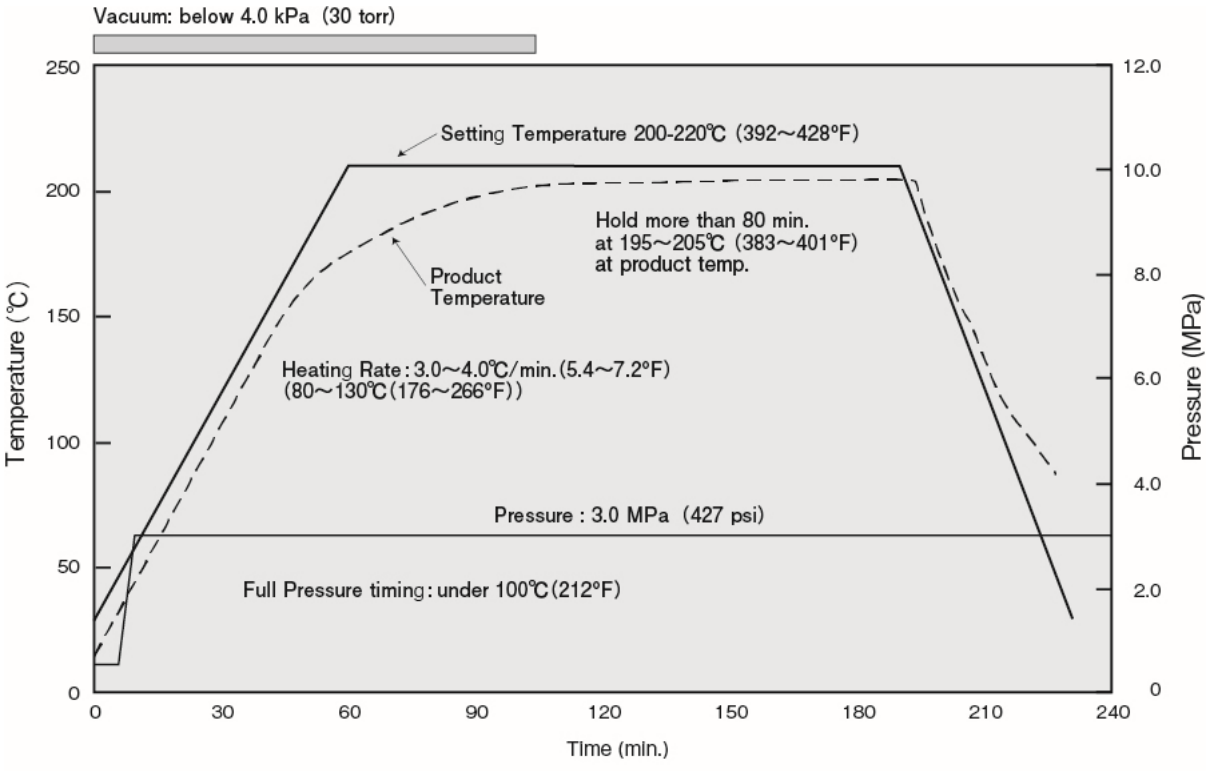
● Precautions for Outer Layer Copper Foil

Some copper foils have low peel strengths and are not suitable.
Please check copper peel strengths of the outer layer copper foil before use.

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.0～4.0℃/min. within the melting temperature range of the prepreg resin (80℃～).
②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.
③The degree of vacuum shall be below 4.0 kPa (30 torr). Please start the vacuuming before adding heat and pressure for lamination.
④The time for vacuuming shall not be longer than the end of molding process. If the vacuuming is continued until the end of the lamination process, flowed resin becomes fragile due to bubble remain.Fragments of it may become a cause of dent. Also small bubbles (blur) may occur in the edge of the product.
⑤The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～4.0 MPa at vacuum pressing.)
⑥We recommend that pressure on material is the low pressure of 0.5 MPa for resin of prepreg at the initial stage and material temperature pressurizes it befor 80℃. The starting pressure is 0.5 MPa, and set the full-pressure, before product temperature become 100℃.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GHA-679G(Type S)

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 3.5～4.5℃/min. within the melting temperature range of the prepreg resin. Hold the product temperature above 210℃ for more than 60 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③The degree of vacuum shall be below 4.0 kPa (30 torr). Please start the vacuuming before adding heat and pressure for lamination.

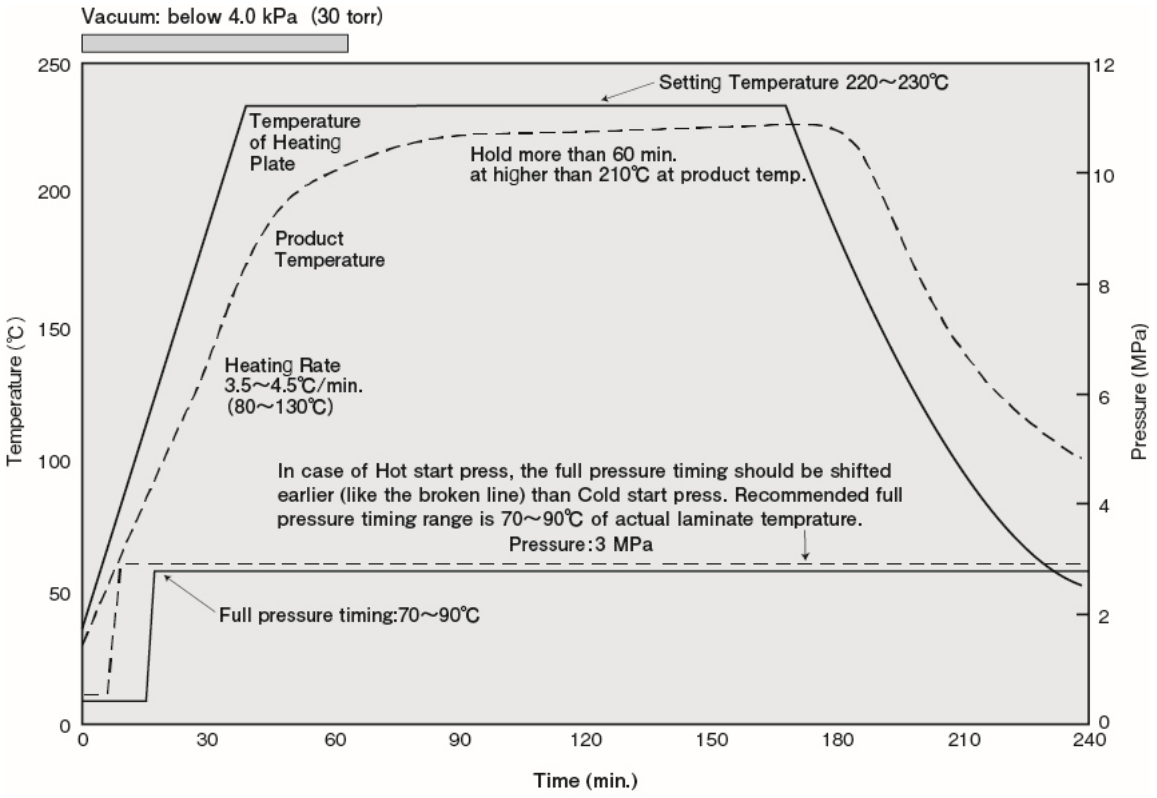
④The time for vacuuming shall not be longer than the end of molding process. If the vacuuming is continued until the end of the lamination process, flowed resin becomes fragile due to bubble remain. Fragments of it may become a cause of dent. Also small bubbles (blur) may occur in the edge of the product.

⑤The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (Standard pressure is 2.5～3.5 MPa at vacuum pressing.)

⑥In order to reduce pressure shock to the product, it is recommended to set initial pressure low, 0.5 MPa, although one-step pressure is basically no problem.

⑦In case of Hot start press, the full pressure timing should be shifted earlier (like the broken line) than Cold start press. Recommended full pressure timing range is 70～90℃ of actual laminate temprature.

● Recommended Press Conditions



※The standard heating profile is for general reference only. Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

GIA-671N

● Precautions for Lamination of Multilayer Board

①Heating rate of the product should be about 1.5～2.5℃/min. within the melting temperature range of the prepreg resin (80～130℃). Hold the product temperature above 170℃ for more than 40 min.

②The setting temperature of the heating plates differs, depending on the heating rate or cushions, so it is necessary to adjust the temprature in order to satisfy the above condition ①.

③For high Tg (over 200℃) MLB, please set the temperature of the heating plates higher than the standard temperature.

④The degree of vacuum shall be below 4.0 kPa (30 torr). Please start the vacuuming before adding heat and pressure for lamination.

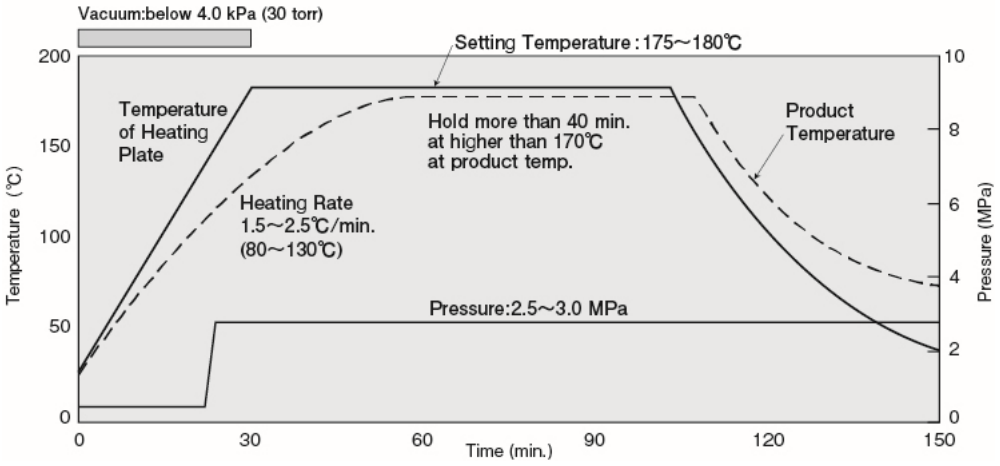
⑤The time for vacuuming shall not be longer than the end of molding process. If the vacuuming is continued until the end of the lamination process, flowed resin becomes fragile due to bubble remain. Fragments of it may become a cause of dent. Also small bubbles (blur) may occur in the edge of the product.

⑥The pressure of the product may influence molding property, thickness between layers or dimensional change, so it is necessary to examine sufficiently to decide the condition. (standard pressure is 2.5～3.0 MPa.)

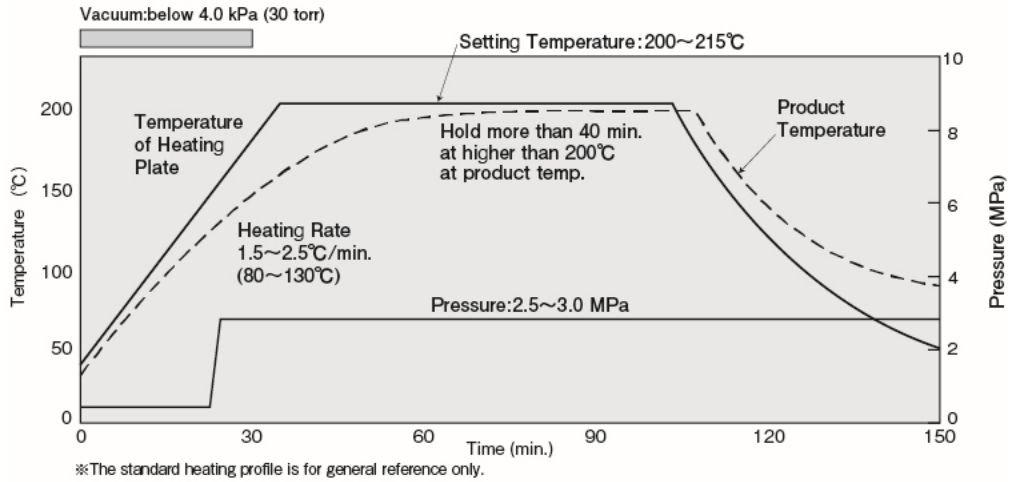
⑦In order to reduce pressure shock to the product, it is recommended to set initial pressure low, 0.5 MPa for 20～30 min., although one-step pressure is basically no problem.

● Recommended Press Conditions

For Standard Multilayer Application



For Multilayer Application Requiring Tg of More than 200℃



※The standard heating profile is for general reference only.
Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

TD-002

● Precautions for Lamination of Multilayer Board

①Heating rate of the product measured from 80～130℃ should be set between 2.0～3.0℃/minutes.The heating time when the product temperature is 175℃ or higher should be 60minutes or longer.

②Setting temperature of heating plates differs depending on the heating rate of the product and the type of cushions used, so it is necessary to adjust the temperature in order to satisfy the above condition ①.

③Vacuuming should be started before heating and pressing the product and degree of vacuum should be below 4.0 kPa（30 torr）.

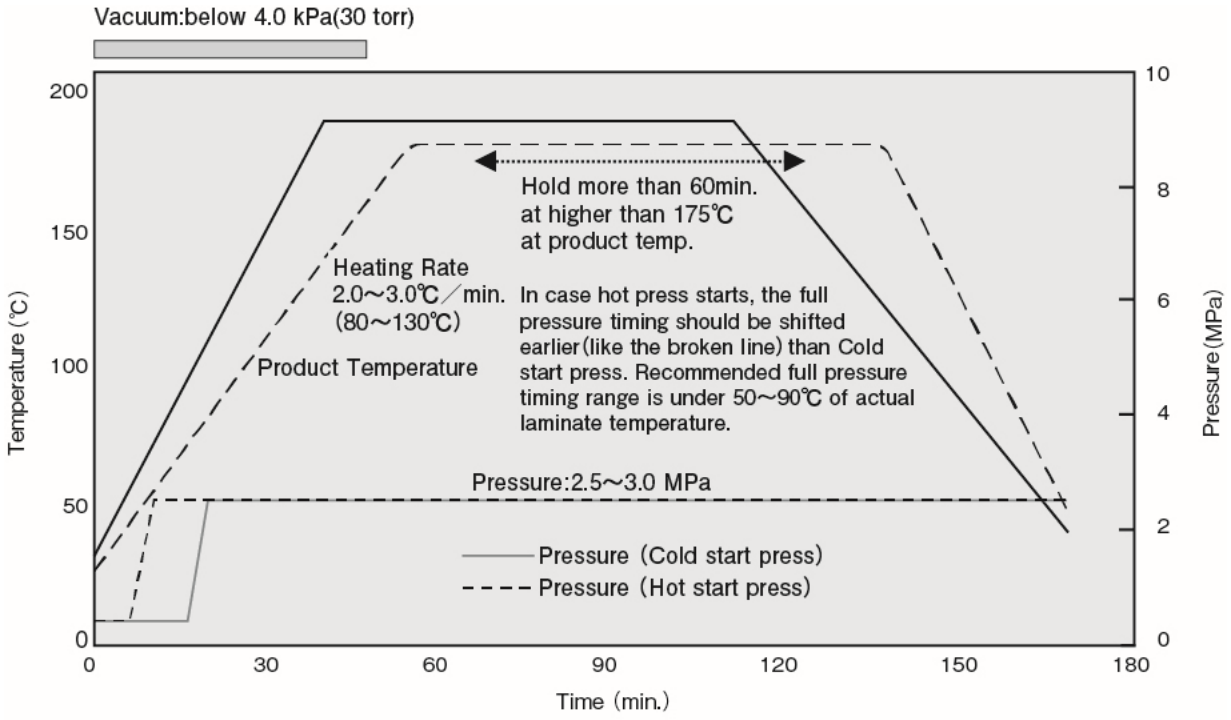
④Vacuuming should not be continued until the end of molding process. If the vacuuming is continued until the end of the lamination process, bubbles might remain in the flowed prepreg resin. As a result, the flowed resin might become fragile and break into fragments. The fragments of resin scattered on the product might cause dents or defaults on the surface of the product. Also fine bubbles and blur might occur at the edge of the product, if the vacuuming is continued till the end.

⑤Lamination pressure should be thoroughly examined, because it might influence the thickness of the product and the rate of dimensional change of the product.（Standard pressure for vacuum pressing is 2.5～3.0 MPa.）

⑥To lessen the pressure shock of the product, two-stage press is recommended. The lamination pressure at the first stage recommended is 0.5 MPa.

⑦In case hot press starts, the full pressure timing should be shifted earlier（ like the broken line）than cold start press. Recommended full pressure timing range is under 50～90℃ of actual laminate temperature.

● Recommended Press Conditions



※The standard heating profile is for general reference only.
Conditions may change depending on metal plate thickness, number of cushion layers, number of MLB lay-up, etc.

Standards UL Standard

● Base Material Recognition File No. E80148

Part Number	ANSI	Min. Thickness	Max. Thickness	UL 94 Flame	TI*1		PLC				DSR
					Elec	Mech	HWI*1	HAI*1	HVTR*1	CTI	
MCL-E-679F	FR-4.0	0.03	1.6	V- 0	130	140	0	1	0	1	YES
MCL-E-679FJ	FR-4.0	0.03	1.6	V- 0	130	140	0	0	—	3	YES
MCL-E-679FG	FR-4.1	0.03	1.6	V- 0,VTM-0	130	140	0	2	0	0	YES
MCL-HE-679G	FR-4.1	0.03	1.6	V- 0,VTM-0	130	140	0	0	—	2	YES
MCL-I-671	GPY	0.03	1.6	V- 0	170	180	0	0	3	4	YES
MCL-E-700G	—	0.02	0.62	V- 0,VTM-0	180	180	3	0	—	0	YES
MCL-E-770G	—	0.10*2	0.81	V- 0	170	140	0	0	—	1	YES
MCL-E-795G	—	0.02	1.6	V- 0,VTM-0	160	160	0	0	—	0	YES
MCL-LW-900G MCL-LW-910G	—	0.05	1.6	V- 0,VTM-0	160	140	0	1	—	1	YES
MCL-LW-990	—	0.04	1.6	V- 0,VTM-0	50	50	0	0	—	1	YES
MCL-HS100	—	0.05	1.6	V- 0,VTM-0	160	140	0	1	—	1	YES
TD-002	—	0.04	1.6	V- 0,VTM-0	50	50	0	1	—	3	YES

*1)Data above are based on unclad laminates, and the properties may vary depending on the board thickness.
TI:Temperature Index(℃) HWI:Hot Wire Ignition(PLC) HAI:High-current Arc Ignition(PLC) HVTR:High-voltage-arc Tracking Rate(PLC) CTI:Comparative Tracking Index(PLC)
*2)Min. laminate and prepreg sheet thicknesses in multilayer PWBs are 0.02 mm.

Hot Wire Ignition Performance Level Categories

Range - Mean Ignition Time (Sec.)	Assigned PLC
120 ≦ IT	0
60 ≦ IT < 120	1
30 ≦ IT < 60	2
15 ≦ IT < 30	3
7 ≦ IT < 15	4
0 ≦ IT < 7	5

High-current Arc Ignition Performance Level Categories

Range - Mean Number of Arcs to Cause Ignition (NA)	Assigned PLC
120 ≦ NA	0
60 ≦ NA < 120	1
30 ≦ NA < 60	2
15 ≦ NA < 30	3
0 ≦ NA < 15	4

High-voltage Arc-tracking-rate Performance Level Categories

Range - Tracking Rate (mm/min.)	Assigned PLC
0 < TR ≦ 10	0
10 < TR ≦ 25.4	1
25.4 < TR ≦ 80	2
80 < TR ≦ 150	3
150 < TR	4

Comparative Tracking Performance Level Categories

Range - Tracking Index (Voltage)	Assigned PLC
600 ≦ TI	0
400 ≦ TI < 600	1
250 ≦ TI < 400	2
175 ≦ TI < 250	3
100 ≦ TI < 175	4
0 ≦ TI < 100	5

Please refer to UL Product iQ for details.

● MCIL Recognition File No. E80148

Part Number	ANSI	Min. Thickness (mm)	UL Flame Class	Clad Conductive Thickness (μm)		Max. Area Diameter (mm)	Solder Limits		Assembly Solder Process(*)				Max. Operating Temperature (℃)
				Min.	Max.		Temperature (℃)	Time (sec.)	Solder Process		Solder Limits		
									Temp.(℃)	Cycles	Temp.(℃)	Time(sec.)	
MCL-E-679F	FR-4.0	0.12	V-0	14.3	35	50.8	*1		—	—	—	—	110
		0.38	V-0	5	70	50.8	*1		—	—	—	—	130
		0.20	V-0	3	35	50.8	—		260	6	260	60	120
		0.30	V-0	3	70	50.8	—		260	6	260	60	120
		0.38	V-0	3	70	50.8	—		260	6	260	60	130
MCL-E-679FJ	FR-4.0	0.20	V-0	3	35	50.8	—		260	6	288	30	130
		0.30	V-0	3	70	50.8	—		260	6	288	30	130
MCL-E-679FG	FR-4.1	0.12	V-0	3	35	50.8	*1		—	—	—	—	110
		0.20	V-0	5	70	50.8	*1		—	—	—	—	125
		0.38	V-0	5	70	50.8	*1		—	—	—	—	130
MCL-HE-679G	FR-4.1	0.17	V-0	3	35	50.8	*3		—	—	—	—	120
		0.20	V-0	3	35	50.8	*3		—	—	—	—	130
		0.38	V-0	3	70	50.8	*3		—	—	—	—	130
		0.12	V-0	3	35	50.8	—		260	6	288	30	120
		0.22	V-0	3	35	50.8	—		260	6	288	30	130
		0.38	V-0	3	70	50.8	—		260	6	288	30	130
MCL-I-671	GPY	0.20	V-0	5	105	50.8	*2		—	—	—	—	130

Certified condition may vary depending on the board thickness.
＊1: 230℃/40 min. + 250℃/40 sec. + 260℃/20 sec. ＊2: 230℃/60 min. + 260℃/2 min. + 260℃/20 sec. ＊3: 230℃/40 min. + 250℃/2 min. + 288℃/30 sec.
(＊)Assembly solder process evaluated to IPC-TM-650, 2.6.27 Thermal Stress Assembly Simulation.

Standards BS Standard

Certificate No.	Part Number	Flammability category	Min. Thickness
VC643585	MCL-I-671	V-0	0.10 mm
VC660376	MCL-E-679F	V-0	0.06 mm
VC656656	MCL-E-679FG	V-0	0.10 mm

Conditions Note

eg. C-96/20/65 + D-2/100
Treat the sample under the Constant temperature and humidity, temperature 20℃, relative humidity 65% for 96 hrs first, then, dip it 100℃ water for 2 hrs.

1. Each alphabet describes sample treatment type.
   A: As received, no treatment applied.
   C: Treat the sample under Constant temperature and humidity
   D: Dip treatment in homothermal water

2. First figure describes treatment time (h)

3. 2nd figure describes treatment temperature (℃)

4. 3rd figure describes treatment relative humidity (%)

5. Alphabets and figures are separated by horizontal line and diagonal line separates the treatment types.

6. + (plus) indicates that 2 and more treatments are conducted in that order.

Notice

- The contents of this brochure are as of October 2023. Product specifications or appearance may be changed without notice.
- The contents of this brochure are based on the test results conducted by Resonac. Values and usage of products are not guaranteed. Please evaluate the products at your site for actual results and use this brochure as a reference.
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Resonac Corporation

Tokyo Shiodome Bldg., 1-9-1, Higashi-Shimbashi, Minato-ku, Tokyo 105-7325, Japan Laminate Materials Business Sector https://www.resonac.com/

- AFFILIATES

Resonac America, Inc.
2150 North First Street, Suite 350, San Jose, CA 95131, U.S.A. Phone: +1-408-873-2200

Resonac Europe GmbH
Daimlerstrasse 19, 86368 Gersthofen, Germany Phone : +49-8271-4243-2600

Resonac (China) Corporation
Unit 1901. Lee Garden. Shanghai No. 668, Xinzha Road, Jing An District, 200041, China Phone : +86-21-6288-8870

Resonac(China) Corporation Beijing Branch
Room 418, Beijing Fortune Bldg, 5 Dongsanhuan (N) Road, Chao Yang District, Beijing, 10004, China Phone : +86-10-6590-8760

Resonac(China) Corporation Shenzhen Branch
Unit 801 West Block, Coastal Building, Nanshan District, Shenzhen, 518054, China Phone : +86-755-8207-0543

Resonac(China) Corporation Chengdu Branch
Suite 2, Floor 11, Tower 2, Chengdu IFS, No.1 Section 3 HongXingRoad, Chengdu, Sichuan Province, 61011, China Phone : +86-28-8459-2221

Resonac International (Taiwan) Co., Ltd.
6F, No.96, Sec. 2, Chung Shan N. Rd., Chung Shan Dist., Taipei City 104, Taiwan Phone : +886-2-2581-3632

Resonac Asia Pacific Pte.Ltd.
4 Shenton Way, #16-02/06, SGX Centre Ⅱ, Singapore 068807 Phone : +65-6836-6988

Resonac Korea Corporation
(Samjung Green Core The City)201-501~519, Dongtangwangyeokhwanseung-ro, 62, Hwaseong-si, Gyeonggi-do, 18479, Korea Phone: +82-31-8047-3700

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